Efficient methods for *ab initio* modeling of materials using finite-element DFT with τ-mGGA functionals

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Abstract

Meta-GGA functionals such as r^{2}SCAN improve markedly on GGA accuracy, but in plane-wave codes their dependence on the kinetic energy density makes each self-consistent iteration several times more expensive. We formulate and implement spin-polarized τ-mGGA Kohn-Sham DFT for both ONCV and PAW pseudopotentials in the GPU-accelerated real-space finite-element code DFT-FE. We validate the energies against Quantum ESPRESSO, and the forces and stresses against the corresponding expressions built from numerical derivatives of the energy. The τ-dependent terms need no FFT-like transformations, and the τ-mGGA overhead over GGA is very small. DFT-FE reaches an 11 to 15 times shorter minimum wall time per iteration than Quantum ESPRESSO on identical architecture. This enables large-scale simulations beyond the reach of plane-wave codes, shown for oxygen vacancies in $BaTiO_3$, CO adsorption on Pt nanoparticles and flat bands in twisted bilayer MoS_2, the largest with 3786 atoms and 32812 electrons.

Introduction

Kohn-Sham density functional theory (KS-DFT) [1,2] is the most widely used framework for first-principles calculations of the electronic structure of materials [3,4]. It gives access to ground-state energies, equilibrium structures and response properties at a cost that scales cubically with the number of electrons. Its accuracy is determined by the approximation to the exchange-correlation (XC) energy functional, whose successive refinements are classified along the rungs of Jacob's ladder [5]. The local density approximation (LDA) [6] and the generalized gradient approximation (GGA) [7] form the two lowest rungs. Their semilocal form makes them cheap to evaluate, and they remain the functionals of choice for large-scale calculations despite their systematic errors. GGA typically overestimates lattice constants [8] and equilibrium volumes [9], and both rungs give an unbalanced description of covalent, ionic and weakly bound interactions. These errors propagate to defect formation energies, surface and adsorption energetics, and the relative stability of competing phases. Hybrid functionals and correlated wave function methods reduce them substantially, but their nonlocal form raises the cost so steeply that these methods are limited to systems far smaller than many applications require.

The third rung, between GGA and the hybrid functionals, comprises meta-generalized gradient approximations that depend additionally on the kinetic energy density τ, denoted τ-mGGA in what follows. Because τ is built from the occupied Kohn-Sham orbitals, it discriminates between covalent, metallic and weak bonding regions, which the density and its gradient alone cannot distinguish. A τ-mGGA functional nevertheless keeps a semilocal form, and hence a cost that is in principle comparable to that of GGA[10]. The strongly constrained and appropriately normed (SCAN) functional[11] was constructed to satisfy the known exact constraints that a semilocal form can meet, and it established that this extra ingredient improves accuracy systematically across diverse bonding environments[12]. SCAN is, however, numerically demanding, since its XC potential and its derivatives show oscillatory structure and its total energy converges slowly as the integration grid is refined[13,14]. These problems motivated the regularized variant rSCAN[13] and then r^{2}SCAN[15], which restores the exact constraints relaxed by the regularization while keeping the smooth numerical behavior of rSCAN. Assessments over large materials datasets have since established the accuracy of r^{2}SCAN for the formation energies, thermodynamic stability, lattice parameters and magnetic properties of solids[16–19].

The orbital dependence responsible for this accuracy is also the source of the difficulties in implementing τ-mGGA functionals. Since the XC energy depends on the orbitals through τ, its variational derivative is not a multiplicative potential. The XC part of the Kohn-Sham Hamiltonian instead contains a differential operator acting on the orbital gradients[20,21]. The atomic forces and unit cell stresses likewise acquire τ-dependent terms that involve the orbital gradients. The numerical integration is also more demanding, because the XC energy density depends on τ through dimensionless variables such as the iso-orbital indicator, which can vary sharply in space. Computed quantities are therefore more sensitive to the accuracy of the quadrature than in the GGA case, and plane-wave codes need substantially denser fast Fourier transform (FFT) grids for τ-mGGA functionals than for GGA[22]. The operator term takes further FFTs, so each application of the Hamiltonian in a plane-wave code needs about four times as many FFTs as in the GGA case. As a result, a τ-mGGA calculation in our plane-wave benchmarks costs roughly two to five times as much per self-consistent field (SCF) iteration as a GGA calculation, far from the near-parity that the semilocal form suggests.

This overhead adds to a more fundamental limit on the system size that plane-wave pseudopotential codes[23–25] can treat. These codes remain the standard for periodic systems, and their basis is systematically convergent but globally supported. The distributed FFTs that apply the Hamiltonian therefore involve all-to-all communication, which limits the MPI concurrency that can be used productively. The memory needed to store the orbitals as plane waves also limits the number of electrons, and non-periodic systems such as nanoparticles and molecules need vacuum padding, which enlarges the basis further. Localized-orbital and linear-scaling methods[26–28] reduce the cost per SCF iteration, but their bases are not systematically convergent, and their localization approximations are hard to control for metallic systems. Finite-difference[29], wavelet[30] and finite-element (FE)[31–34] discretizations combine the strengths of both approaches. Their basis functions are compactly supported, and refining the grid or mesh reduces the discretization error systematically, for any atomic species and configuration.

Among these, the FE method is particularly well suited to large-scale KS-DFT calculations on GPU-accelerated machines. Its basis of piecewise higher-order polynomials controls the discretization error through two independent parameters, the mesh size and the polynomial order. The mesh can also be refined adaptively near the nuclei and coarsened away from them, so that the number of basis functions stays small for a given accuracy. The real space framework treats periodic, non-periodic and semi-periodic boundary conditions without taking the recourse to vacuum padding. Because the basis functions are compactly supported, the discrete Hamiltonian is sparse, and applying it needs communication only between neighboring subdomains. Within each finite element, the Hamiltonian is applied through dense linear algebra of high arithmetic intensity, which maps efficiently onto GPUs. These properties are exploited in the open-source DFT-FE code (https://sites.google.com/umich.edu/dftfe)[35,36], which has sustained up to 659.7 PFLOPS on the Frontier supercomputer in mixed-precision simulations of quasicrystals and interacting extended defects in metallic alloys with a machine-learned XC functional[37]. DFT-FE was a finalist for the 2019 ACM Gordon Bell Prize[38] and the winner in 2023. The code also incorporates Chebyshev-filtered subspace iteration and SCF acceleration techniques[39–41], as well as an FE-discretized projector augmented-wave (PAW) formulation[42].

In this work, we develop a spin-polarized τ-mGGA formulation within the higher-order FE discretized, optimized norm-conserving Vanderbilt (ONCV) pseudopotential[43] framework of DFT-FE. For the atomic forces and periodic unit cell stresses, we extend the generalized (configurational) force approach[44–46]

to τ-mGGA, which gives both quantities from a single expression. Finally, we devise a strategy for the finer integration that the τ-mGGA integrands require, at a cost that does not depend on the number of electronic states. The integration accuracy can therefore be raised as needed, without any approximation beyond the FE discretization.

We validate the implementation against the plane-wave code Quantum ESPRESSO (QE)[24,25] and show that, in DFT-FE, the ratio of τ-mGGA to GGA cost per SCF iteration stays near unity. We then demonstrate that τ-mGGA calculations are practical for large-scale applications through three problems that need several thousand atoms. These are the neutral oxygen vacancy in $BaTiO_3$ in the dilute limit, CO adsorption on Pt nanoparticles of increasing size and flat bands in twisted bilayer MoS_2 at small twist angles.

The strategy developed here rests on the properties of the real space FE discretization rather than on the treatment of the core region. It therefore carries over directly to the PAW-FE framework developed within DFT-FE[42] and to its GPU-accelerated extension[47]. In both, the smooth PAW pseudo-densities admit coarser meshes and markedly fewer basis functions than the norm-conserving setting at comparable accuracy. Our PAW-FE τ-mGGA implementation reproduces the cost trends reported here for the ONCV case, with the ratio of τ-mGGA to GGA cost again near unity. The near-parity between τ-mGGA and GGA is therefore a property of the FE discretization, not of the pseudopotential formalism. Together with the smaller basis and the demonstrated scalability of PAW-FE to about a hundred thousand electrons[47], this near-parity places τ-mGGA accuracy within reach of systems appreciably larger than those considered here. A detailed account of the PAW-FE τ-mGGA implementation and the associated benchmarks is given in the Supplementary Information.

Results

We first present the τ-mGGA formulation and its finite-element implementation in DFT-FE, then benchmark its accuracy and performance against QE, and finally apply it to three large-scale problems.

Mathematical formulation

In this section, we discuss the mathematical framework underlying τ-mGGA functionals within the KS-DFT formalism. We begin with the spin-polarized Kohn-Sham energy functional in the τ-mGGA formalism, identifying the additional orbital dependence introduced through τ relative to LDA and GGA. The formulation is cast within the optimized norm-conserving Vanderbilt (ONCV) pseudopotential framework, wherein the core electrons are replaced by smooth pseudopotentials, retaining only the valence electron degrees of freedom. The corresponding τ-mGGA formulation within the projector augmented-wave (PAW) framework is presented in Supplementary Section S1. We then discuss the τ-mGGA contributions to the Kohn-Sham Hamiltonian. Finally, we extend the generalized force approach[44,46] to KS-DFT with τ-mGGA functionals. The generalized force approach takes the derivative of the ground-state energy with respect to a deformation of the underlying space. A deformation localized around an atom gives the force on that atom, and an affine deformation of the cell gives the stress. Forces and stresses thus follow from a single expression. The dependence of the XC energy on τ adds terms to this expression that are absent in the GGA case.

DFT governing equations with τ-mGGA functional

For a periodic crystalline system, Bloch's theorem dictates that the single-electron wave functions take the form $\psi^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}) = e^{i\boldsymbol{k}\cdot\mathbf{x}}\, u^{\sigma}_{n\boldsymbol{k}}(\mathbf{x})$, where $u^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}+\mathbf{L}) = u^{\sigma}_{n\boldsymbol{k}}(\mathbf{x})$ for all lattice vectors $\mathbf{L}$, with the crystal momentum $\boldsymbol{k}$ restricted to the first Brillouin zone (BZ). We denote the periodic unit cell by Ω_p and $\fint_{\mathrm{BZ}}$ denotes the integration over the BZ of the periodic unit cell Ω_p, normalized by the BZ volume. For non-periodic systems, the BZ integration reduces to a single $\boldsymbol{k} = \mathbf{0}$ point, Ω_p is replaced by $\mathbb{R}^3$ (in practice, a sufficiently large domain with Dirichlet boundary conditions), and the $\boldsymbol{k}$ label is dropped. The equations that follow are written in the periodic notation, with the non-periodic case recovered as the appropriate limit.

The ground-state properties of a system comprising N_a nuclei and N_e electrons are governed by KS-DFT[2]. In the pseudopotential setting adopted here, the core electrons are replaced by pseudopotentials and only the N_v valence electrons are treated explicitly. In the spin-polarized τ-mGGA formalism, the

ground state is obtained by minimizing the energy functional

$$E[\{\psi^\sigma_{n\boldsymbol{k}}\},\{\mathbf{R}_a\}] = \min_{\{\psi^\sigma_{n\boldsymbol{k}}\}} \{T_s + E_{\rm xc} + E_{\rm el} + E_{\rm psp} + E_{\rm ent}\} \quad \text{s.t.} \quad \langle\psi^\sigma_{n\boldsymbol{k}}|\psi^{\sigma'}_{n'\boldsymbol{k}'}\rangle = \delta_{nn'}\delta_{\boldsymbol{k}\boldsymbol{k}'}\delta_{\sigma\sigma'}, \tag{1}$$

where $\{\psi^\sigma_{n\boldsymbol{k}}\}$ are the single-electron wave functions with spin index $\sigma \in \{\uparrow,\downarrow\}$, band index $1 \le n \le N$ ($N \ge N_v/2$), and $\{\mathbf{R}_a\}$ denotes the nuclear positions of the N_a atoms. The non-interacting kinetic energy T_s and the XC energy $E_{\rm xc}$ are respectively given by

$$T_s[\{\psi^\sigma_{n\boldsymbol{k}}\}] = \sum_\sigma \sum_n \fint_{\rm BZ} \int_{\Omega_p} f^\sigma_{n\boldsymbol{k}}\, \psi^{\sigma *}_{n\boldsymbol{k}} \left(-\frac{1}{2}\nabla^2\right) \psi^\sigma_{n\boldsymbol{k}}\, d\mathbf{x}\, d\boldsymbol{k}, \tag{2}$$

$$E_{\rm xc}[\{\rho^\sigma_{\rm xc}\},\{\nabla\rho^\sigma_{\rm xc}\},\{\tau^\sigma_{\rm xc}\}] = \int_{\Omega_p} e_{\rm xc}\big(\{\rho^\sigma_{\rm xc}\},\,\{\nabla\rho^\sigma_{\rm xc}\},\,\{\tau^\sigma_{\rm xc}\}\big)\, d\mathbf{x}, \tag{3}$$

where $f^\sigma_{n\boldsymbol{k}} \equiv f(\epsilon^\sigma_{n\boldsymbol{k}},\mu)$ is the Fermi-Dirac occupation function and μ is the chemical potential. Unlike LDA and GGA functionals[5–7], which depend solely on the spin-resolved densities ρ^σ and their gradients $\nabla\rho^\sigma$, the τ-mGGA XC energy density $e_{\rm xc}$ [11,15] additionally depends on the spin-resolved kinetic energy densities τ^σ, introducing an explicit orbital dependence into the functional. The ρ^σ and τ^σ are computed from the Kohn-Sham wave functions as

$$\rho^\sigma(\mathbf{x}) = \sum_n \fint_{\rm BZ} f^\sigma_{n\boldsymbol{k}}\, |\psi^\sigma_{n\boldsymbol{k}}(\mathbf{x})|^2\, d\boldsymbol{k}, \qquad \tau^\sigma(\mathbf{x}) = \frac{1}{2}\sum_n \fint_{\rm BZ} f^\sigma_{n\boldsymbol{k}}\, |\nabla\psi^\sigma_{n\boldsymbol{k}}(\mathbf{x})|^2\, d\boldsymbol{k}. \tag{4}$$

In Eq. (3), $e_{\rm xc}$ is evaluated at the nonlinear core corrected (NLCC) spin-resolved fields $\rho^\sigma_{\rm xc}(\mathbf{x}) := \rho^\sigma(\mathbf{x}) + \frac{1}{2}\rho_{\rm c}(\mathbf{x})$ and $\tau^\sigma_{\rm xc}(\mathbf{x}) := \tau^\sigma(\mathbf{x}) + \frac{1}{2}\tau_{\rm c}(\mathbf{x})$, where

$$\rho_{\rm c}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_r \rho^a_{\rm c}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|), \qquad \tau_{\rm c}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_r \tau^a_{\rm c}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|) \tag{5}$$

are the atom-centered core electron density and core kinetic energy density tabulated by the pseudopotential, and the sums over r run over the lattice translation vectors $\mathbf{L}_r$. For pseudopotentials without NLCC, one sets $\rho_{\rm c} = \tau_{\rm c} = 0$, recovering $\rho^\sigma_{\rm xc} \to \rho^\sigma$ and $\tau^\sigma_{\rm xc} \to \tau^\sigma$.

The local real space reformulation of $E_{\rm el}$ differs from refs. 35,36 by using a compact reference charge $b(\mathbf{x}) = \sum_a b^a(\mathbf{x}-\mathbf{R}_a)$ and a separate auxiliary smeared nuclear charge $g(\mathbf{x}) = \sum_a g^a(\mathbf{x}-\mathbf{R}_a)$, with $\int b^a(\mathbf{x}-\mathbf{R}_a)d\mathbf{x} = \int g^a(\mathbf{x}-\mathbf{R}_a)d\mathbf{x} = -Z^v_a$. Here, $Z^v_a > 0$ denotes the valence charge of atom a. The sums defining b and g include periodic images. Both charge distributions use the normalized radial profile

$$\eta(r;R) = \begin{cases} \dfrac{21(R-r)^3(6r^2+3rR+R^2)}{5\pi R^8}, & 0 \le r \le R, \\ 0, & r > R, \end{cases} \tag{6}$$

with $b^a(\mathbf{x}-\mathbf{R}_a) = -Z^v_a\eta(|\mathbf{x}-\mathbf{R}_a|;R_b)$ and $g^a(\mathbf{x}-\mathbf{R}_a) = -Z^v_a\eta(|\mathbf{x}-\mathbf{R}_a|;R^a_g)$. The reference radius R_b is chosen so that the compact charges do not overlap. The auxiliary charges may overlap and are used in the electrostatic Poisson problem. The electrostatic energy takes the form

$$E_{\rm el}[\rho,\{\mathbf{R}_a\}] = \frac{1}{2}\int_{\Omega_p} (\rho+g)\, V_{\rm el}\, d\mathbf{x} - C_{\rm el}(\{\mathbf{R}_a\}), \tag{7}$$

where $\rho(\mathbf{x}) = \rho^\uparrow(\mathbf{x}) + \rho^\downarrow(\mathbf{x})$ is the total electron density, and $V_{\rm el}$ is the electrostatic potential satisfying the Poisson equation

$$-\frac{1}{4\pi}\nabla^2 V_{\rm el} = \rho + g. \tag{8}$$

For a periodic neutral cell, $\int_{\Omega_p}(\rho+g)d\,\mathbf{x} = 0$, and the additive constant in $V_{\rm el}$ is fixed by $\int_{\Omega_p} V_{\rm el}\, d\mathbf{x} = 0$. For an isolated system, the fields are taken to decay at infinity. Subtracting the analytic correction $C_{\rm el}$ in

Eq. (7) removes the self-energies of g and corrects the interactions between overlapping auxiliary charges. The remaining ion–ion energy is that of point charges. The correction is given by

$$C_{\mathrm{el}}(\{\mathbf{R}_a\}) = \frac{1}{2}\int_{\Omega_p} g\, V_g\, d\mathbf{x} - \left(\frac{1}{2}\int_{\Omega_p} b\, V_b\, d\mathbf{x} - E_{\mathrm{self},b}\right), \tag{9}$$

where $V_g(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_r V_g^a(\mathbf{x}, \mathbf{R}_a + \mathbf{L}_r)$ and $V_b(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_r V_b^a(\mathbf{x}, \mathbf{R}_a + \mathbf{L}_r)$. The potentials V_g^a and V_b^a are generated by g^a and b^a, respectively, and satisfy the Poisson equations

$$-\frac{1}{4\pi}\nabla^2 V_g^a(\mathbf{x}, \mathbf{R}_a) = g^a(\mathbf{x} - \mathbf{R}_a), \qquad -\frac{1}{4\pi}\nabla^2 V_b^a(\mathbf{x}, \mathbf{R}_a) = b^a(\mathbf{x} - \mathbf{R}_a). \tag{10}$$

With $r = |\mathbf{x} - \mathbf{R}_a|$ and $R = R_g^a$, each V_g^a is evaluated analytically as

$$V_g^a(\mathbf{x}, \mathbf{R}_a) = -Z_a^v \begin{cases} \dfrac{9r^7 - 30r^6R + 28r^5R^2 - 14r^2R^5 + 12R^7}{5R^8}, & r \le R, \\ 1/r, & r > R. \end{cases} \tag{11}$$

The self-energy of the compact reference charges is

$$E_{\mathrm{self},b} = \frac{1}{2}\sum_{a\in\Omega_p}\int_{\mathbb{R}^3} b^a(\mathbf{x} - \mathbf{R}_a) V_b^a(\mathbf{x}, \mathbf{R}_a)\, d\mathbf{x}. \tag{12}$$

For the profiles in Eq. (6), define the unit-charge pair interaction

$$\mathcal{K}_{aa'}(d) = \int_{\mathbb{R}^3}\int_{\mathbb{R}^3} \frac{\eta(|\mathbf{x}|; R_g^a)\eta(|\mathbf{x}'|; R_g^{a'})}{|\mathbf{x} - \mathbf{x}' - \mathbf{d}|}\, d\mathbf{x}'\, d\mathbf{x}, \qquad |\mathbf{d}| = d. \tag{13}$$

Then Eq. (9) is evaluated analytically as

$$C_{\mathrm{el}} = \frac{1}{2}\sum_{a\in\Omega_p}(Z_a^v)^2\mathcal{K}_{aa}(0) + \frac{1}{2}\sum_{a,a'\in\Omega_p}{\sum_r}' Z_a^v Z_{a'}^v \left[\mathcal{K}_{aa'}(d_{aa'r}) - \frac{1}{d_{aa'r}}\right], \tag{14}$$

where $d_{aa'r} = |\mathbf{R}_a - \mathbf{R}_{a'} - \mathbf{L}_r|$, the prime excludes $a = a'$ with $\mathbf{L}_r = \mathbf{0}$, and $\mathcal{K}_{aa}(0) = 31924/(17875R_g^a)$. The pair interactions and their derivatives are evaluated by exact integration of the piecewise polynomials. For $d \ge R_g^a + R_g^{a'}$, $\mathcal{K}_{aa'}(d) = 1/d$, so only overlapping auxiliary charges contribute to the pair correction. If $g = b$, Eq. (7) reduces to the usual smeared-charge expression with the nuclear self-energy subtracted. [4,36]

Within the ONCV pseudopotential formalism [43], the pseudopotential contribution E_{psp} decomposes into local and nonlocal parts, $E_{\mathrm{psp}} = E_{\mathrm{loc}} + E_{\mathrm{nloc}}$, where

$$E_{\mathrm{loc}}[\rho] = \int_{\Omega_p} (V_{\mathrm{loc}} - V_g)\, \rho\, d\mathbf{x}. \tag{15}$$

Here $V_{\mathrm{loc}}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_r V_{\mathrm{loc}}^a(|\mathbf{x} - \mathbf{R}_a - \mathbf{L}_r|)$ is the superposition of atom-centered local pseudopotentials. The subtraction of V_g in Eq. (15) cancels the electron interaction with g already included in E_{el}. The nonlocal pseudopotential energy is given by

$$E_{\mathrm{nloc}}[\{\psi_{n\boldsymbol{k}}^\sigma\}, \{\mathbf{R}_a\}] = \sum_\sigma\sum_n \fint_{\mathrm{BZ}}\int_{\Omega_p}\int_{\mathbb{R}^3} f_{n\boldsymbol{k}}^\sigma\, \psi_{n\boldsymbol{k}}^{\sigma *}(\mathbf{x})\, V_{\mathrm{nloc}}(\mathbf{x}, \mathbf{x}')\, \psi_{n\boldsymbol{k}}^\sigma(\mathbf{x}')\, d\mathbf{x}'\, d\mathbf{x}\, d\boldsymbol{k}, \tag{16}$$

where the action of the nonlocal operator on a wave function is given by

$$\begin{aligned}\int_{\mathbb{R}^3} V_{\mathrm{nloc}}(\mathbf{x}, \mathbf{x}')\, \psi_{n\boldsymbol{k}}^\sigma(\mathbf{x}')\, d\mathbf{x}' = &\sum_{a\in\Omega_p}\sum_r\sum_{\beta,\beta'} p_\beta^a(\mathbf{x} - (\mathbf{R}_a + \mathbf{L}_r))\, D_{\beta\beta'}^a \\ &\times \int_{\mathbb{R}^3} p_{\beta'}^a(\mathbf{x}' - (\mathbf{R}_a + \mathbf{L}_r))\, \psi_{n\boldsymbol{k}}^\sigma(\mathbf{x}')\, d\mathbf{x}'.\end{aligned} \tag{17}$$

Here β is a composite index containing the radial projector label and the orbital and magnetic angular momentum quantum numbers l and m. Further, $p^a_\beta(\mathbf{x}-\mathbf{R}_a)$ are the atom-centered projector functions and $D^a_{\beta\beta'}$ are the pseudopotential coupling coefficients.

The entropic energy contribution E_{ent} in Eq. (1), arising from the use of fractional Fermi-Dirac occupations, is given by

$$E_{\text{ent}}[\{f^\sigma_{n\boldsymbol{k}}\}] = k_B T \sum_\sigma \sum_n \fint_{\text{BZ}} \left[f^\sigma_{n\boldsymbol{k}} \ln f^\sigma_{n\boldsymbol{k}} + (1-f^\sigma_{n\boldsymbol{k}}) \ln (1-f^\sigma_{n\boldsymbol{k}}) \right] d\boldsymbol{k}, \tag{18}$$

where k_B is the Boltzmann constant, T is the electronic smearing temperature, and the chemical potential μ is determined by enforcing $\sum_\sigma \sum_n \fint_{\text{BZ}} f^\sigma_{n\boldsymbol{k}}\, d\boldsymbol{k} = N_v$.

Combining all contributions, the ground-state problem can be cast as the following variational form

$$\begin{aligned} E[\{\psi^\sigma_{n\boldsymbol{k}}\},\{\mathbf{R}_a\}] = \min_{\{\psi^\sigma_{n\boldsymbol{k}}\}} \Bigg\{ & T_s[\{\psi^\sigma_{n\boldsymbol{k}}\}] + E_{\text{xc}}[\{\rho^\sigma\},\{\nabla\rho^\sigma\},\{\tau^\sigma\}] + \frac{1}{2}\int_{\Omega_p} (\rho+g)\, V_{\text{el}}\, d\mathbf{x} \\ & - C_{\text{el}}(\{\mathbf{R}_a\}) + E_{\text{loc}}[\rho] + E_{\text{nloc}}[\{\psi^\sigma_{n\boldsymbol{k}}\}] + E_{\text{ent}} \Bigg\}. \end{aligned} \tag{19}$$

Minimization of Eq. (19) subject to the orthonormality constraint $\langle \psi^\sigma_{n\boldsymbol{k}} | \psi^{\sigma'}_{n'\boldsymbol{k}'} \rangle = \delta_{nn'}\delta_{\boldsymbol{k}\boldsymbol{k}'}\delta_{\sigma\sigma'}$ yields the Hermitian eigenvalue problem $\mathcal{H}^\sigma_{\boldsymbol{k}} \psi^\sigma_{n\boldsymbol{k}} = \epsilon^\sigma_{n\boldsymbol{k}} \psi^\sigma_{n\boldsymbol{k}}$ for the Kohn-Sham Hamiltonian $\mathcal{H}^\sigma_{\boldsymbol{k}}$, to be solved for the lowest $N \geq N_v/2$ eigenpairs for each $\boldsymbol{k}$ in the first BZ and each spin channel σ. For periodic systems, invoking Bloch's theorem $\psi^\sigma_{n\boldsymbol{k}}(\mathbf{x}) = e^{i\boldsymbol{k}\cdot\mathbf{x}} u^\sigma_{n\boldsymbol{k}}(\mathbf{x})$ transforms the eigenvalue problem for the periodic part $u^\sigma_{n\boldsymbol{k}}$ into

$$\mathcal{H}^\sigma_{\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}} = \epsilon^\sigma_{n\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}}, \quad \text{with} \quad \mathcal{H}^\sigma_{\boldsymbol{k}} = \mathcal{H}^\sigma_{\text{loc},\boldsymbol{k}} + \mathcal{H}^\sigma_{\text{nloc},\boldsymbol{k}}. \tag{20}$$

The local part of the Hamiltonian is

$$\mathcal{H}^\sigma_{\text{loc},\boldsymbol{k}} = -\frac{1}{2}\nabla^2 - i\boldsymbol{k}\cdot\nabla + \frac{1}{2}|\boldsymbol{k}|^2 + V^\sigma_{\text{eff}}, \tag{21}$$

where the spin-dependent effective potential is $V^\sigma_{\text{eff}} = V_{\text{el}} + V_{\text{loc}} - V_g + \hat{V}^\sigma_{\text{xc}}$, and the XC operator $\hat{V}^\sigma_{\text{xc}}$ acts on $u^\sigma_{n\boldsymbol{k}}$ as

$$\begin{aligned} \hat{V}^\sigma_{\text{xc}}\, u^\sigma_{n\boldsymbol{k}} = & \left[\frac{\partial e_{\text{xc}}}{\partial \rho^\sigma_{\text{xc}}} - \nabla\cdot\left(\frac{\partial e_{\text{xc}}}{\partial(\nabla\rho^\sigma_{\text{xc}})} \right) \right] u^\sigma_{n\boldsymbol{k}} \\ & - \frac{1}{2}\left[\nabla\cdot\left(\frac{\partial e_{\text{xc}}}{\partial \tau^\sigma_{\text{xc}}} \nabla u^\sigma_{n\boldsymbol{k}} \right) + \nabla\cdot\left(\frac{\partial e_{\text{xc}}}{\partial \tau^\sigma_{\text{xc}}} i\boldsymbol{k}\, u^\sigma_{n\boldsymbol{k}} \right) - |\boldsymbol{k}|^2 \frac{\partial e_{\text{xc}}}{\partial \tau^\sigma_{\text{xc}}} u^\sigma_{n\boldsymbol{k}} + \frac{\partial e_{\text{xc}}}{\partial \tau^\sigma_{\text{xc}}} i\boldsymbol{k}\cdot\nabla u^\sigma_{n\boldsymbol{k}} \right]. \end{aligned} \tag{22}$$

The first two terms in Eq. (22) form the multiplicative XC potential, which has the same form as in GGA, while the remaining terms, arising from the τ-dependence of e_{xc}, contain non-multiplicative operators that act directly on the orbital, a distinguishing feature of the τ-mGGA formalism. The nonlocal part $\mathcal{H}^\sigma_{\text{nloc},\boldsymbol{k}}$ acts on $u^\sigma_{n\boldsymbol{k}}$ as

$$\begin{aligned} \mathcal{H}^\sigma_{\text{nloc},\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}}(\mathbf{x}) := & \sum_{a\in\Omega_p} \sum_{\beta,\beta'} \left\{ \sum_r e^{-i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)}\, p^a_\beta(\mathbf{x}-(\mathbf{R}_a+\mathbf{L}_r)) \right\} D^a_{\beta\beta'} \\ & \times \left\{ \int_{\Omega_p} \sum_{r'} e^{i\boldsymbol{k}\cdot(\mathbf{x}'-\mathbf{L}_{r'})}\, p^a_{\beta'}(\mathbf{x}'-(\mathbf{R}_a+\mathbf{L}_{r'}))\, u^\sigma_{n\boldsymbol{k}}(\mathbf{x}')\, d\mathbf{x}' \right\}, \end{aligned} \tag{23}$$

where the independent sums over r and r' run over all lattice translation vectors $\mathbf{L}_r$ and $\mathbf{L}_{r'}$, accounting for the interaction of the Bloch wave function with all periodic images of the projectors.

Collecting the above, the ONCV τ-mGGA ground-state problem reduces to the following coupled set of nonlinear equations, to be solved self-consistently for ρ^σ and τ^σ

$$\left(-\frac{1}{2}\nabla^2 - i\boldsymbol{k}\cdot\nabla + \frac{1}{2}|\boldsymbol{k}|^2 + V_{\mathrm{el}} + V_{\mathrm{loc}} - V_g + \hat{V}_{\mathrm{xc}}^{\sigma} + \mathcal{H}_{\mathrm{nloc},\boldsymbol{k}}^{\sigma}\right) u_{n\boldsymbol{k}}^{\sigma} = \epsilon_{n\boldsymbol{k}}^{\sigma}\, u_{n\boldsymbol{k}}^{\sigma},$$

$$1 \leq n \leq N, \qquad \forall\, \boldsymbol{k} \in \mathrm{BZ}, \qquad \forall\, \sigma \in \{\uparrow, \downarrow\}$$

$$\hat{V}_{\mathrm{xc}}^{\sigma}\, u_{n\boldsymbol{k}}^{\sigma} = \left[\frac{\partial e_{\mathrm{xc}}}{\partial \rho_{\mathrm{xc}}^{\sigma}} - \nabla\cdot\left(\frac{\partial e_{\mathrm{xc}}}{\partial(\nabla\rho_{\mathrm{xc}}^{\sigma})}\right)\right] u_{n\boldsymbol{k}}^{\sigma}$$
$$-\frac{1}{2}\left[\nabla\cdot\left(\frac{\partial e_{\mathrm{xc}}}{\partial \tau_{\mathrm{xc}}^{\sigma}}\,\nabla u_{n\boldsymbol{k}}^{\sigma}\right) + \nabla\cdot\left(\frac{\partial e_{\mathrm{xc}}}{\partial \tau_{\mathrm{xc}}^{\sigma}}\, i\boldsymbol{k}\, u_{n\boldsymbol{k}}^{\sigma}\right) - |\boldsymbol{k}|^2\,\frac{\partial e_{\mathrm{xc}}}{\partial \tau_{\mathrm{xc}}^{\sigma}}\, u_{n\boldsymbol{k}}^{\sigma} + \frac{\partial e_{\mathrm{xc}}}{\partial \tau_{\mathrm{xc}}^{\sigma}}\, i\boldsymbol{k}\cdot\nabla u_{n\boldsymbol{k}}^{\sigma}\right]$$

$$\mathcal{H}_{\mathrm{nloc},\boldsymbol{k}}^{\sigma}\, u_{n\boldsymbol{k}}^{\sigma}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_{\beta,\beta'}\left\{\sum_{r} e^{-i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)}\, p_{\beta}^{a}(\mathbf{x} - (\mathbf{R}_a + \mathbf{L}_r))\right\} D_{\beta\beta'}^{a}$$
$$\times\left\{\int_{\Omega_p}\sum_{r'} e^{i\boldsymbol{k}\cdot(\mathbf{x}'-\mathbf{L}_{r'})}\, p_{\beta'}^{a}(\mathbf{x}' - (\mathbf{R}_a + \mathbf{L}_{r'}))\, u_{n\boldsymbol{k}}^{\sigma}(\mathbf{x}')\, d\mathbf{x}'\right\}$$

$$\rho^{\sigma} = \sum_{n} \fint_{\mathrm{BZ}} f_{n\boldsymbol{k}}^{\sigma}\, |u_{n\boldsymbol{k}}^{\sigma}|^2\, d\boldsymbol{k}, \qquad \rho = \rho^{\uparrow} + \rho^{\downarrow}$$

$$\tau^{\sigma} = \frac{1}{2}\sum_{n} \fint_{\mathrm{BZ}} f_{n\boldsymbol{k}}^{\sigma}\left[\left|\nabla u_{n\boldsymbol{k}}^{\sigma}\right|^2 + |\boldsymbol{k}|^2\left|u_{n\boldsymbol{k}}^{\sigma}\right|^2 - i\boldsymbol{k}\cdot(u_{n\boldsymbol{k}}^{\sigma *}\nabla u_{n\boldsymbol{k}}^{\sigma} - u_{n\boldsymbol{k}}^{\sigma}\nabla u_{n\boldsymbol{k}}^{\sigma *})\right] d\boldsymbol{k}$$

$$\rho_{\mathrm{xc}}^{\sigma} = \rho^{\sigma} + \frac{1}{2}\rho_{\mathrm{c}}, \qquad \tau_{\mathrm{xc}}^{\sigma} = \tau^{\sigma} + \frac{1}{2}\tau_{\mathrm{c}}$$

$$-\frac{1}{4\pi}\nabla^2 V_{\mathrm{el}} = \rho + g, \qquad -\frac{1}{4\pi}\nabla^2 V_g^a(\mathbf{x}, \mathbf{R}_a) = g^a(\mathbf{x} - \mathbf{R}_a),\ V_g = \sum_{a\in\Omega_p}\sum_{r} V_g^a(\mathbf{x}, \mathbf{R}_a + \mathbf{L}_r)$$

$$f_{n\boldsymbol{k}}^{\sigma} = \left[1 + \exp\left(\frac{\epsilon_{n\boldsymbol{k}}^{\sigma} - \mu}{k_B T}\right)\right]^{-1}, \qquad \sum_{\sigma}\sum_{n} \fint_{\mathrm{BZ}} f_{n\boldsymbol{k}}^{\sigma}\, d\boldsymbol{k} = N_v$$

Generalized force approach for τ-mGGA functionals

To evaluate atomic forces and unit cell stresses, we employ the generalized force approach which was originally proposed by Eshelby[48] in the context of continuum mechanics. It was subsequently adapted to KS-DFT by Motamarri and Gavini[44] and further extended to noncollinear magnetism and spin-orbit coupling by Kodali and Motamarri[46]. In the τ-mGGA formalism, the additional dependence of e_{xc} on τ^{σ} introduces new contributions to these expressions that are absent in the GGA case.

Following the approach of Kodali and Motamarri[46], we introduce a generalized energy functional within the spin-polarized τ-mGGA setting

$$\begin{aligned}\widetilde{E}[\{\widetilde{V}^{\sigma}\},\{\widetilde{V}_{\tau}^{\sigma}\},\{\widetilde{\rho}^{\sigma}\},\{\widetilde{\tau}^{\sigma}\},\{\widetilde{f}_{n\boldsymbol{k}}^{\sigma}\}] = &\sum_{\sigma}\sum_{n} \fint_{\mathrm{BZ}} \widetilde{f}_{n\boldsymbol{k}}^{\sigma}\, \epsilon_{n\boldsymbol{k}}^{\sigma}[\widetilde{V}^{\sigma}, \widetilde{V}_{\tau}^{\sigma}]\, d\boldsymbol{k} - \sum_{\sigma}\int_{\Omega_p}\left(\widetilde{V}^{\sigma}\widetilde{\rho}^{\sigma} + \widetilde{V}_{\tau}^{\sigma}\widetilde{\tau}^{\sigma}\right) d\mathbf{x} \\ &+ E_{\mathrm{el}}[\widetilde{\rho}] + E_{\mathrm{xc}}[\{\widetilde{\rho}^{\sigma} + \tfrac{1}{2}\rho_{\mathrm{c}}\}, \{\widetilde{\tau}^{\sigma} + \tfrac{1}{2}\tau_{\mathrm{c}}\}] \\ &+ E_{\mathrm{ent}}[\{\widetilde{f}_{n\boldsymbol{k}}^{\sigma}\}] + \mu\left(N_v - \sum_{\sigma}\sum_{n} \fint_{\mathrm{BZ}} \widetilde{f}_{n\boldsymbol{k}}^{\sigma}\, d\boldsymbol{k}\right),\end{aligned} \tag{24}$$

where the potentials $\widetilde{V}^\sigma$ and $\widetilde{V}^\sigma_\tau$, the densities $\widetilde{\rho}^\sigma$ and $\widetilde{\tau}^\sigma$, and the occupations $\widetilde{f}^\sigma_{n\boldsymbol{k}}$ are independent variables, and $\epsilon^\sigma_{n\boldsymbol{k}}[\widetilde{V}^\sigma, \widetilde{V}^\sigma_\tau]$ are the eigenvalues of the Hamiltonian defined by

$$\begin{aligned}
&\widetilde{\mathcal{H}}^\sigma_{\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}} = \epsilon^\sigma_{n\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}}, \quad \widetilde{\mathcal{H}}^\sigma_{\boldsymbol{k}} = \widetilde{\mathcal{H}}^\sigma_{\mathrm{loc},\boldsymbol{k}} + \widetilde{\mathcal{H}}^\sigma_{\mathrm{nloc},\boldsymbol{k}},\\
&\widetilde{\mathcal{H}}^\sigma_{\mathrm{loc},\boldsymbol{k}} = -\frac{1}{2}\nabla^2 - i\boldsymbol{k}\cdot\nabla + \frac{|\boldsymbol{k}|^2}{2} + \bar{V}^\sigma + \hat{V}^\sigma_{\tau,\mathrm{xc}}[\widetilde{V}^\sigma_\tau], \quad \bar{V}^\sigma = \widetilde{V}^\sigma + V_{\mathrm{loc}} - V_g,
\end{aligned} \tag{25}$$

with $\hat{V}^\sigma_{\tau,\mathrm{xc}}[\widetilde{V}^\sigma_\tau]$ acting on $u^\sigma_{n\boldsymbol{k}}$ as

$$\hat{V}^\sigma_{\tau,\mathrm{xc}}\, u^\sigma_{n\boldsymbol{k}} = -\frac{1}{2}\Big[\nabla\cdot\Big(\widetilde{V}^\sigma_\tau\,\nabla u^\sigma_{n\boldsymbol{k}}\Big) + \nabla\cdot\Big(\widetilde{V}^\sigma_\tau\, i\boldsymbol{k}\, u^\sigma_{n\boldsymbol{k}}\Big) - |\boldsymbol{k}|^2\,\widetilde{V}^\sigma_\tau\, u^\sigma_{n\boldsymbol{k}} + \widetilde{V}^\sigma_\tau\, i\boldsymbol{k}\cdot\nabla u^\sigma_{n\boldsymbol{k}}\Big]. \tag{26}$$

The nonlocal contribution $\widetilde{\mathcal{H}}^\sigma_{\mathrm{nloc},\boldsymbol{k}}$ has the same form as defined in Eq. (23). Setting the functional derivatives of $\widetilde{E}$ with respect to $\widetilde{V}^\sigma$, $\widetilde{V}^\sigma_\tau$, $\widetilde{\rho}^\sigma$, $\widetilde{\tau}^\sigma$, and $\widetilde{f}^\sigma_{n\boldsymbol{k}}$ to zero recovers the self-consistent Kohn-Sham equations with NLCC-augmented XC potentials, confirming that this functional has a stationary point, denoted $\widetilde{E}_S$, at the Kohn-Sham ground state (see Supplementary Section S2 for the proof).

To compute the generalized force, we consider a parametrized perturbation Γ^ε of the reference configuration, $\mathbf{x}^\varepsilon = \Gamma^\varepsilon(\mathbf{x})$, with generator $\boldsymbol{\Upsilon} = d\Gamma^\varepsilon/d\varepsilon|_{\varepsilon=0}$. By evaluating the directional derivative of $\widetilde{E}^\varepsilon_S$, the stationary value of $\widetilde{E}$ in the perturbed configuration, at $\varepsilon = 0$ and exploiting the vanishing of all variational derivatives at the stationary point (see Supplementary Section S3 for the detailed derivation and Supplementary Section S4 for the NLCC contribution), the generalized force expression takes the form

$$\left.\frac{d\widetilde{E}^\varepsilon_S}{d\varepsilon}\right|_{\varepsilon=0} = \int_{\Omega_p} (\mathbf{E} + \mathbf{E}_\tau) : \nabla\boldsymbol{\Upsilon}\, d\mathbf{x} + F^{\mathrm{psp,nloc}} + F^K + F^{\mathrm{ext,corr}} + F^{\mathrm{sm}} + F^C + F^{\mathrm{NLCC}}, \tag{27}$$

where ":" denotes the double contraction of two rank-2 tensors and $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3$. The tensors $\mathbf{E}_2$ and $\mathbf{E}_3$ are the electrostatic and XC contributions, respectively, while $F^{\mathrm{psp,nloc}}$, $F^{\mathrm{ext,corr}}$, F^{sm}, and F^C capture the nonlocal pseudopotential, local pseudopotential, smooth nuclear charge, and analytic correction contributions. The tensor $\mathbf{E}_2$ and these four force terms do not involve τ^σ, and $\mathbf{E}_3$ depends on it only through e_{xc} and $\partial e_{\mathrm{xc}}/\partial(\nabla\rho^\sigma_{\mathrm{xc}})$. All six terms therefore keep their GGA form, and their spin-polarized expressions are collected in Supplementary Section S3. The remaining terms, $\mathbf{E}_1$, $\mathbf{E}_\tau$, F^K, and F^{NLCC}, carry the additional τ-dependence of the τ-mGGA functional and are described next.

The rank-2 Eshelby-like tensor $\mathbf{E}_1$ is modified in the τ-mGGA case to include the τ-dependent terms

$$\begin{aligned}
\mathbf{E}_1 = &\left(\sum_\sigma\sum_n \fint_{\mathrm{BZ}} \frac{f^\sigma_{n\boldsymbol{k}}}{2}\Big[\nabla u^{\sigma*}_{n\boldsymbol{k}}\cdot\nabla u^\sigma_{n\boldsymbol{k}} + \left(|\boldsymbol{k}|^2 - 2\epsilon^\sigma_{n\boldsymbol{k}}\right) u^{\sigma*}_{n\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}} - 2i\, u^{\sigma*}_{n\boldsymbol{k}}\,\boldsymbol{k}\cdot\nabla u^\sigma_{n\boldsymbol{k}}\Big]\, d\boldsymbol{k} + (V_{\mathrm{loc}} - V_g)\,\rho\right)\boldsymbol{I}\\
&- \sum_\sigma\sum_n \fint_{\mathrm{BZ}} \frac{f^\sigma_{n\boldsymbol{k}}}{2}\left(1 + \frac{\partial e_{\mathrm{xc}}}{\partial\tau^\sigma_{\mathrm{xc}}}\right)\Big[\nabla u^{\sigma*}_{n\boldsymbol{k}}\otimes\nabla u^\sigma_{n\boldsymbol{k}} + \nabla u^\sigma_{n\boldsymbol{k}}\otimes\nabla u^{\sigma*}_{n\boldsymbol{k}} - 2i\, u^{\sigma*}_{n\boldsymbol{k}}\,\nabla u^\sigma_{n\boldsymbol{k}}\otimes\boldsymbol{k}\Big]\, d\boldsymbol{k},
\end{aligned} \tag{28}$$

where $\boldsymbol{I}$ is the identity tensor and $\otimes$ denotes the outer product. The purely τ-mGGA contribution $\mathbf{E}_\tau$ has the expression

$$\mathbf{E}_\tau = -\sum_\sigma\sum_n \fint_{\mathrm{BZ}} i\,\frac{f^\sigma_{n\boldsymbol{k}}}{2}\,\frac{\partial e_{\mathrm{xc}}}{\partial\tau^\sigma_{\mathrm{xc}}}\left(u^\sigma_{n\boldsymbol{k}}\,\nabla u^{\sigma*}_{n\boldsymbol{k}} + u^{\sigma*}_{n\boldsymbol{k}}\,\nabla u^\sigma_{n\boldsymbol{k}}\right)\otimes\boldsymbol{k}\, d\boldsymbol{k}. \tag{29}$$

The k-point-dependent force contributions, also modified by the τ-mGGA functional, are

$$\begin{aligned}
F^K = &\sum_\sigma\sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}} \int_{\Omega_p}\left(1 + \frac{\partial e_{\mathrm{xc}}}{\partial\tau^\sigma_{\mathrm{xc}}}\right)\left(-i\, u^{\sigma*}_{n\boldsymbol{k}}\,\frac{\partial\boldsymbol{k}^\varepsilon}{\partial\varepsilon}\bigg|_{\varepsilon=0}\cdot\nabla u^\sigma_{n\boldsymbol{k}} + \frac{1}{2}\,\frac{\partial|\boldsymbol{k}^\varepsilon|^2}{\partial\varepsilon}\bigg|_{\varepsilon=0} u^{\sigma*}_{n\boldsymbol{k}}\, u^\sigma_{n\boldsymbol{k}}\right) d\mathbf{x}\, d\boldsymbol{k}\\
&+ \sum_\sigma\sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}} \int_{\Omega_p}\frac{i}{2}\,\frac{\partial e_{\mathrm{xc}}}{\partial\tau^\sigma_{\mathrm{xc}}}\,\frac{\partial\boldsymbol{k}^\varepsilon}{\partial\varepsilon}\bigg|_{\varepsilon=0}\cdot\left(u^\sigma_{n\boldsymbol{k}}\,\nabla u^{\sigma*}_{n\boldsymbol{k}} + u^{\sigma*}_{n\boldsymbol{k}}\,\nabla u^\sigma_{n\boldsymbol{k}}\right) d\mathbf{x}\, d\boldsymbol{k} + F^{\mathrm{psp,nloc}}_{\boldsymbol{k}},
\end{aligned} \tag{30}$$

where $\boldsymbol{k}^\varepsilon$ is the wave vector after the deformation Γ^ε, and $F^{\mathrm{psp,nloc}}_{\boldsymbol{k}}$ denotes the $\boldsymbol{k}$-dependent part of the nonlocal pseudopotential force, which is unchanged from the GGA case[46].

Since ρ_c and τ_c translate rigidly with the nuclei, e_{xc} acquires an explicit $\{\mathbf{R}_a\}$-dependence that contributes an additional atom-centered force $F^{\mathrm{NLCC}} = F_\rho^{\mathrm{NLCC}} + F_\tau^{\mathrm{NLCC}}$ from the core density and the core kinetic energy density, given in Supplementary Section S4.

Atomic forces are obtained from Eq. (27) with a generator localized about each atom. The unit cell stress is defined as the derivative of the ground-state energy with respect to the homogeneous strain, divided by the unit cell volume,

$$\sigma_{ij} = \frac{1}{|\Omega_p|}\frac{\partial E_0}{\partial \varepsilon_{ij}}, \tag{31}$$

where $E_0 = \widetilde{E}_S$ is the ground-state energy and ε_{ij} are the components of the strain. The stress follows from Eq. (27) with the affine generator $\boldsymbol{\Upsilon}(\mathbf{x}) = \boldsymbol{C}\mathbf{x}$, where $\boldsymbol{C}$ is a constant 3×3 matrix. The resulting expressions for the atomic forces and the unit cell stress, in the form used in the implementation, are given in Supplementary Section S5. They reduce to the GGA expressions when $\partial e_{xc}/\partial \tau_{xc}^\sigma = 0$ and $\tau_c = 0$.

Numerical implementation

The mathematical formulation presented above is implemented within the open-source DFT-FE framework[35,36], which employs higher order finite-element discretization for large-scale KS-DFT calculations. In this section, we describe the key numerical aspects of the implementation. We first present the finite-element discretization of the governing equations, including the discrete Hamiltonian matrix incorporating the τ-dependent terms and the efficient solution of the resulting generalized eigenvalue problem using a Chebyshev filtered subspace iteration procedure. Subsequently, we describe an efficient strategy for evaluating the spin-resolved electron density, its gradient, and τ^σ at the quadrature points, which is central to achieving near-GGA computational cost at the τ-mGGA level.

Finite element discretization

In the finite-element (FE) method, the simulation domain Ω_p is decomposed into non-overlapping hexahedral cells by generating an FE mesh (triangulation) $\mathcal{T}^h$ with characteristic mesh size h. The underlying basis functions are compactly supported, piecewise C^0 continuous polynomials constructed as tensor products of 1D Lagrange polynomials of degree p defined over Gauss-Lobatto-Legendre (GLL) nodal points[34–36]. The discrete periodic Bloch wave functions are expanded in this basis as follows,

$$u_{n\boldsymbol{k}}^{\sigma,h}(\mathbf{x}) = \sum_{I=1}^{M^{h,p}} N_I^{h,p}(\mathbf{x})\, u_{n\boldsymbol{k},I}^{\sigma}, \tag{32}$$

where $N_I^{h,p}$ are the $M^{h,p}$ FE basis functions and $u_{n\boldsymbol{k},I}^{\sigma}$ are the complex linear combination coefficients. Substituting Eq. (32) into the eigenvalue problem Eq. (20) and applying a Galerkin projection yields the generalized Hermitian eigenvalue problem (GHEP)

$$\mathbf{H}_{\boldsymbol{k}}^{\sigma}\, \hat{\boldsymbol{u}}_{n\boldsymbol{k}}^{\sigma} = \epsilon_{n\boldsymbol{k}}^{\sigma,h}\, \mathbf{M}\, \hat{\boldsymbol{u}}_{n\boldsymbol{k}}^{\sigma}, \qquad M_{IJ} = \int_{\Omega_p} N_I^{h,p}\, N_J^{h,p}\, d\mathbf{x}, \tag{33}$$

where $\hat{\boldsymbol{u}}_{n\boldsymbol{k}}^{\sigma}$ is the vector of expansion coefficients, $\mathbf{H}_{\boldsymbol{k}}^{\sigma} = \mathbf{H}_{\mathrm{loc},\boldsymbol{k}}^{\sigma} + \mathbf{H}_{\mathrm{nloc},\boldsymbol{k}}^{\sigma}$ is the $M^{h,p} \times M^{h,p}$ FE-discretized Hamiltonian matrix, and $\mathbf{M}$ is the FE overlap matrix. The local part of the discrete Hamiltonian is obtained by projecting the local Hamiltonian operator of Eq. (21), together with the XC operator of Eq. (22), onto the FE basis. The divergence terms in Eq. (22) are treated via integration by parts within the Galerkin projection, transferring the derivatives onto the basis functions and thereby avoiding the evaluation of second derivatives of the electronic fields, which require higher regularity than afforded by the C^0 FE basis. The resulting local Hamiltonian matrix has entries

$$\begin{aligned}(\mathbf{H}_{\mathrm{loc},\boldsymbol{k}}^{\sigma})_{IJ} = \int_{\Omega_p} \Bigg[&\frac{1}{2}\left(1 + \frac{\partial e_{xc}^h}{\partial \tau_{xc}^{\sigma,h}}\right) \nabla N_I^{h,p} \cdot \nabla N_J^{h,p} - i N_I^{h,p}\, \boldsymbol{k} \cdot \nabla N_J^{h,p} \\ &+ \left(V_{el}^h + V_{loc} - V_g + \frac{\partial e_{xc}^h}{\partial \rho_{xc}^{\sigma,h}} + \frac{|\boldsymbol{k}|^2}{2}\left(1 + \frac{\partial e_{xc}^h}{\partial \tau_{xc}^{\sigma,h}}\right)\right) N_I^{h,p}\, N_J^{h,p}\end{aligned}$$

$$+ \frac{\partial e_{\mathrm{xc}}^{h}}{\partial(\nabla \rho_{\mathrm{xc}}^{\sigma,h})} \cdot \nabla\left(N_I^{h,p} N_J^{h,p}\right) + \frac{i}{2} \frac{\partial e_{\mathrm{xc}}^{h}}{\partial \tau_{\mathrm{xc}}^{\sigma,h}} \boldsymbol{k} \cdot \left(N_J^{h,p} \nabla N_I^{h,p} - N_I^{h,p} \nabla N_J^{h,p}\right) \Bigg] d\mathbf{x}, \tag{34}$$

where the superscript h marks fields computed from the FE-discretized wave functions. All τ-mGGA contributions to the discrete Hamiltonian enter through $\partial e_{\mathrm{xc}}/\partial \tau_{\mathrm{xc}}^{\sigma}$, and setting this derivative to zero recovers the standard GGA Hamiltonian matrix[36].

The discretized electrostatic potential $V_{\mathrm{el}}^{h}(\mathbf{x}) = \sum_I N_I^{h,p_{\mathrm{el}}}(\mathbf{x})\, V_{\mathrm{el},I}$ appearing in Eq. (34) is constructed using the polynomial order p_{el} on the same FE triangulation $\mathcal{T}^h$ and is obtained by solving the FE-discretized Poisson equation

$$\mathbf{K}\,\hat{\mathbf{V}}_{\mathrm{el}} = \mathbf{c}, \qquad K_{IJ} = \frac{1}{4\pi} \int_{\Omega_p} \nabla N_I^{h,p_{\mathrm{el}}} \cdot \nabla N_J^{h,p_{\mathrm{el}}}\, d\mathbf{x}, \qquad c_I = \int_{\Omega_p} N_I^{h,p_{\mathrm{el}}} (\rho^h + g)\, d\mathbf{x}, \tag{35}$$

where $\hat{\mathbf{V}}_{\mathrm{el}}$ is the vector of coefficients $V_{\mathrm{el},I}$, $\mathbf{K}$ is the stiffness matrix, and $\mathbf{c}$ is the forcing vector. The auxiliary charges are evaluated directly at the quadrature points and normalized per atom so that their quadrature integrals equal $-Z_a^v$. The radii R_g^a are specified directly or chosen using the local pseudopotential tail and FE mesh size. They are limited by the non-periodic boundaries and the range of periodic images included in the calculation. The potential V_g is evaluated using Eq. (11), without an additional FE Poisson solve. At fixed atomic positions and FE mesh, $\mathbf{K}$, V_g, and V_{loc} are precomputed and reused throughout the SCF iterations. The solution for V_{el}^h is recomputed at each SCF iteration using a conjugate-gradient (CG) solver as the electron density ρ^h evolves, and $V_{\mathrm{el}}^h + V_{\mathrm{loc}} - V_g$ is then assembled at the quadrature points for the local Hamiltonian. Each integral over Ω_p above is first split into a sum of integrals over the FE cells $\Omega^{(e)}$. The integral over each cell is then mapped to the reference cell $\hat{\Omega} = [-1,1]^3$ and approximated by Gauss-Legendre quadrature. For an integrand f, we obtain

$$\int_{\Omega_p} f(\mathbf{x})\, d\mathbf{x} = \sum_e \int_{\Omega^{(e)}} f(\mathbf{x})\, d\mathbf{x} = \sum_e \int_{\hat{\Omega}} f\left(\mathbf{x}^{(e)}(\hat{\mathbf{x}})\right) J^{(e)}(\hat{\mathbf{x}})\, d\hat{\mathbf{x}} \approx \sum_e \sum_q w_q\, f(\mathbf{x}_q)\, J^{(e)}(\hat{\mathbf{x}}_q), \tag{36}$$

where $\mathbf{x}^{(e)}(\hat{\mathbf{x}})$ maps $\hat{\Omega}$ onto $\Omega^{(e)}$ and $J^{(e)} = \det\left(\partial \mathbf{x}^{(e)}/\partial \hat{\mathbf{x}}\right)$ is the determinant of its Jacobian. The points $\hat{\mathbf{x}}_q$ and weights w_q belong to a tensor-product Gauss-Legendre rule on $\hat{\Omega}$, and $\mathbf{x}_q = \mathbf{x}^{(e)}(\hat{\mathbf{x}}_q)$ are the corresponding quadrature points in $\Omega^{(e)}$.

The GHEP in Eq. (33) is solved using a Chebyshev filtered subspace iteration (ChFSI) procedure[39]. To efficiently evaluate the action of $\mathbf{M}^{-1}$, required within ChFSI, the GLL quadrature rule coincident with the FE nodal points is employed to construct a diagonal approximation $\mathbf{M}_D$ to $\mathbf{M}$. A Chebyshev polynomial filter of degree s is then applied to the preconditioned Hamiltonian $\bar{\mathbf{H}}_{\boldsymbol{k}}^{\sigma} = \mathbf{M}_D^{-1}\mathbf{H}_{\boldsymbol{k}}^{\sigma}$, which is first scaled and shifted so that the unwanted spectrum $[\lambda_T, \lambda_{\max}]$ maps to $[-1,1]$ while the wanted spectrum $[\lambda_{\min}, \lambda_T)$ lies below -1. The spectral bounds $\lambda_{\min}$ and $\lambda_{\max}$ are estimated via a generalized Lanczos procedure[49], while λ_T is approximated from the highest Rayleigh quotient of $\mathbf{H}_{\boldsymbol{k}}^{\sigma}$ from the previous SCF iteration. The exponential growth of Chebyshev polynomials outside $[-1,1]$ amplifies the desired eigenspace while suppressing the unwanted components, yielding a filtered subspace $\mathbf{X}_s$ rich in the lowest N eigenvectors. A Rayleigh-Ritz projection is then performed by solving the subspace eigenvalue problem $\mathbf{X}_s^{\dagger}\mathbf{H}_{\boldsymbol{k}}^{\sigma}\mathbf{X}_s\,\mathbf{Q}_s = \mathbf{X}_s^{\dagger}\mathbf{M}\mathbf{X}_s\,\mathbf{Q}_s\,\mathbf{\Lambda}_s$, yielding Ritz eigenvectors $\mathbf{U} = \mathbf{X}_s\mathbf{Q}_s$ and Ritz values $\mathbf{\Lambda}_s$. Following Kodali et al.[40], the filtering step is reformulated using Hamiltonian-times-residual products, which progressively decrease in magnitude as the SCF converges. Errors introduced into these products by the diagonal approximation of the inverse $\mathbf{M}_D^{-1}$ shrink with the residual and do not limit the final accuracy. As a result, the computed eigenvalues and eigenvectors converge to those of Eq. (33) with the full overlap matrix $\mathbf{M}$, even though only its diagonal approximation $\mathbf{M}_D$ is used. The same property allows the filtering to be carried out in mixed precision, which reduces the computational cost. This cycle of filtering and Rayleigh-Ritz projection is repeated within each SCF iteration until a desired residual tolerance is achieved.

Efficient evaluation of density, density gradient, and kinetic energy density

The XC energy density e_{xc} and its derivatives appearing in Eq. (34) must be evaluated at the quadrature points from the spin-resolved electron density ρ^{σ}, its gradient $\nabla\rho^{\sigma}$, and τ^{σ}. For τ-mGGA functionals, e_{xc} depends on τ^{σ} through dimensionless variables such as the iso-orbital indicator, which can vary sharply in space. As a result, e_{xc} and its derivatives are harder to integrate accurately than for a GGA[13,14], and

a finer quadrature grid with $Q > 2p+1$ points per direction is needed. A straightforward approach would interpolate the wave functions $u^{\sigma}_{n\boldsymbol{k}}$ and their gradients from the FE nodal points directly to this finer quadrature grid and then evaluate ρ^{σ}, $\nabla\rho^{\sigma}$, and τ^{σ} pointwise. However, this wave function interpolation step scales with the number of wave functions solved and dominates the cost when performed on the finer quadrature grid.

We instead exploit the fact that, within each FE cell, ρ^{σ}, $\nabla\rho^{\sigma}$, and τ^{σ} are polynomials of degree at most $2p$ in each coordinate direction. They can therefore be exactly represented by the Lagrange polynomial basis $\{\bar{N}_q(\mathbf{x})\}$ constructed over the standard quadrature grid with $2p+1$ points per direction as

$$\rho^{\sigma}(\mathbf{x}) = \sum_q \rho^{\sigma}(\mathbf{x}_q)\,\bar{N}_q(\mathbf{x}), \quad \nabla\rho^{\sigma}(\mathbf{x}) = \sum_q \rho^{\sigma}(\mathbf{x}_q)\,\nabla\bar{N}_q(\mathbf{x}), \quad \tau^{\sigma}(\mathbf{x}) = \sum_q \tau^{\sigma}(\mathbf{x}_q)\,\bar{N}_q(\mathbf{x}). \tag{37}$$

Here $\mathbf{x}_q$ are the $(2p+1)^3$ quadrature points of the cell. The values $\rho^{\sigma}(\mathbf{x}_q)$ and $\tau^{\sigma}(\mathbf{x}_q)$ are computed from Eq. (4), after the wave functions and their gradients are interpolated to these points from the $(p+1)^3$ nodes of the cell using Eq. (32). Equation (37) then gives ρ^{σ}, $\nabla\rho^{\sigma}$, and τ^{σ} on the finer quadrature grid required by the τ-mGGA functional. The wave functions are thus never interpolated to that grid. The approach is illustrated schematically in Fig. 1 and introduces no additional approximation beyond the original FE discretization, since the $2p+1$ grid exactly captures the polynomial content of these fields.

In both approaches, the wave functions are interpolated by dense matrix products on each FE cell. Interpolating the N wave functions and their gradients from the $(p+1)^3$ nodes of a cell to its Q^3 points of the finer quadrature grid therefore costs $\mathcal{O}\big(NQ^3(p+1)^3\big)$, and to its $(2p+1)^3$ points $\mathcal{O}\big(N(2p+1)^3(p+1)^3\big)$. Evaluating the sums in Eq. (4) at these points costs less by a factor of $(p+1)^3$. The re-interpolation in Eq. (37) maps the $(2p+1)^3$ values of each field to the Q^3 points at a cost of $\mathcal{O}\big(Q^3(2p+1)^3\big)$ per cell. It acts on ρ^{σ} and τ^{σ} after the sums over the wave functions and the k-points. Its cost therefore depends neither on N nor on the number of k-points $N_{\boldsymbol{k}}$. With N_{cell} FE cells, the costs per spin channel in each SCF iteration are given by

$$C_{\mathrm{direct}} = \mathcal{O}(N_{\boldsymbol{k}}N_{\mathrm{cell}}NQ^3(p+1)^3), \; C_{\mathrm{opt}} = \mathcal{O}(N_{\boldsymbol{k}}N_{\mathrm{cell}}N(2p+1)^3(p+1)^3) + \mathcal{O}(N_{\mathrm{cell}}Q^3(2p+1)^3), \tag{38}$$

for the straightforward and the optimized approach, respectively. The first term of C_{opt} is the cost of computing $\rho^{\sigma}(\mathbf{x}_q)$ and $\tau^{\sigma}(\mathbf{x}_q)$, and the second is the cost of the re-interpolation. The second term is negligible when $N_{\boldsymbol{k}}N \gg \big(Q/(p+1)\big)^3$, which holds in practice since N is in the hundreds or more. The optimized approach then reduces the cost by the factor $\big(Q/(2p+1)\big)^3$.

Accuracy benchmarking

Ground-state energies

We first compare r^{2}SCAN ground-state free energies from DFT-FE with those from the plane-wave code QE for a set of periodic systems (see Methods for the computational settings). The test set contains metallic La and Li and the oxides La_2O_3, Li_2O and ZrO_2, all of which are relevant to solid-state battery applications.

Table 1 Comparison of r^{2}SCAN free energies between QE and DFT-FE.

System	N_{atoms}	QE (Ha)	DFT-FE (Ha)	$\lvert\Delta E\rvert$ (Ha/atom)
La	32	-1.023923×10^3	-1.023924×10^3	5.57×10^{-6}
La_2O_3	5	-1.122895×10^2	-1.122896×10^2	1.29×10^{-5}
Li	16	-1.142437×10^2	-1.142439×10^2	8.30×10^{-6}
Li_2O	12	-1.215962×10^2	-1.215963×10^2	9.00×10^{-6}
ZrO_2	12	-3.186468×10^2	-3.186469×10^2	1.31×10^{-5}

Table 1 lists the free energies from both DFT-FE and QE. For the metals and the insulators alike, the difference is at most 1.3×10^{-5} Ha/atom, well within chemical accuracy ($\sim 1.6 \times 10^{-3}$ Ha).

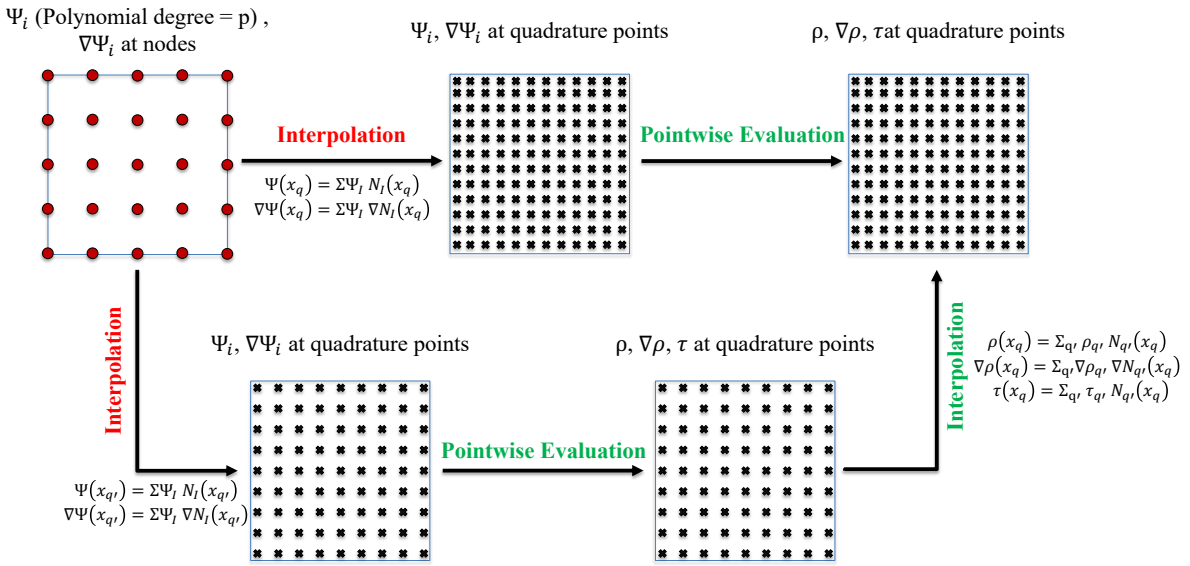


Fig. 1 Schematic illustration of the two strategies for evaluating ρ^σ, $\nabla\rho^\sigma$, and τ^σ at quadrature points. The top panel shows the standard approach using a finer quadrature grid with $Q > 2p+1$ points per direction applied directly to the wave functions (expensive, red). The bottom panel shows the optimized approach adopted in DFT-FE, where ρ^σ, $\nabla\rho^\sigma$, and τ^σ are first evaluated at a $2p+1$ grid and then cheaply interpolated to the finer quadrature grid via a secondary Lagrange basis (green). The two approaches yield identical accuracy while the optimized strategy avoids the expensive wave function interpolation to the finer quadrature grid.

Atomic forces and unit cell stresses

We next assess the atomic forces and unit cell stresses computed with the generalized force approach of Supplementary Eqs. (S5.3) and (S5.12). Both contain τ-dependent terms that are absent in GGA. To verify these terms, we compare the computed values with references obtained by numerically differentiating the energy with respect to the atomic positions (for forces) and the lattice strain (for stresses).[1]

For the force validation, we use a CO molecule. The r^{2}SCAN ground-state energy is computed at several C–O bond lengths, and the negative derivative of a polynomial fit to these energies with respect to the bond length serves as the reference force. Figure 2a compares the computed force on the C atom with this reference. We observe close agreement across all bond lengths considered, confirming the consistency of the force implementation with the energy functional.

For the stress validation, we use the conventional four-atom cubic cell of fcc Al and compute the free energy E at several values of the lattice parameter a. For the isotropic deformation $a^\varepsilon = (1+\varepsilon)a$ of a cubic cell with $|\Omega_p| = a^3$, Eq. (31) gives the diagonal stress component as

$$\sigma = \frac{1}{3|\Omega_p|}\frac{dE}{d\varepsilon} = \frac{1}{3a^2}\frac{dE}{da}. \tag{39}$$

The reference stress is obtained by fitting a polynomial to the computed energies and differentiating it analytically. Figure 2b shows close agreement between the computed stresses and the reference. Since the τ-dependent Eshelby tensor $\mathbf{E}_\tau$ enters the cell stress but not the atomic forces, this test also verifies its contribution.

Performance benchmarking

We now assess the computational performance of DFT-FE with the r^{2}SCAN τ-mGGA functional relative to the plane-wave code QE, using a periodic and a non-periodic benchmark set (see Methods for the benchmark systems, the computational settings, the performance metrics and the hardware).

[1]The τ-mGGA unit cell stress of QE is incorrect in our tests, both in v7.5 and in the latest release v7.6, so the stresses are not compared with QE.

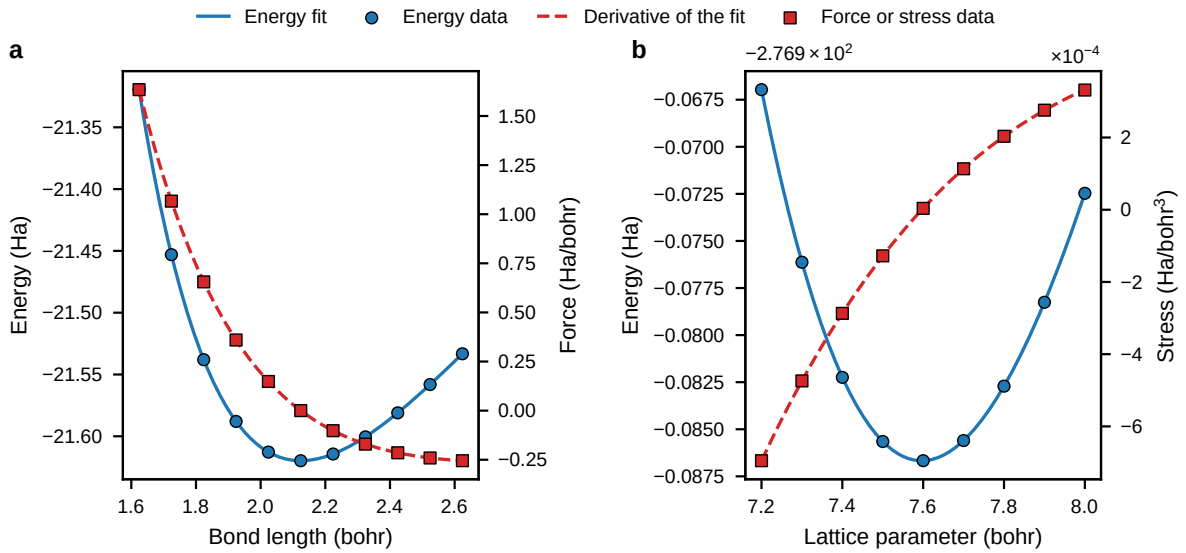


Fig. 2 Validation of the r^{2}SCAN atomic forces and unit cell stresses of DFT-FE against energy derivatives. **a** Ground-state energy and atomic force on the C atom as a function of C–O bond length. **b** Free energy and diagonal stress component σ as a function of lattice parameter for fcc Al. In both panels, the blue circles are the computed energies, the blue line is the polynomial fit to them, the red squares are the computed force (**a**) or stress (**b**), and the red dashed line is the reference obtained from the derivative of the fit.

Computational efficiency comparison

Compared with the GGA case, the τ-mGGA formalism needs τ^σ and adds the non-multiplicative terms of Eq. (22) to the Hamiltonian. Their cost depends strongly on the basis, plane waves in QE and local FE basis functions in DFT-FE. In QE, the extra work comes from the FFTs. Building τ^σ once per SCF iteration needs the gradient of every wave function in real space, which takes three extra inverse FFTs per wave function. In addition, every time the eigensolver applies the Hamiltonian to a trial vector, the τ-dependent term needs the gradient of that vector in real space, multiplied there by $\partial e_{\mathrm{xc}}/\partial \tau^\sigma_{\mathrm{xc}}$ and transformed back. This takes three inverse and three forward FFTs per vector. Each application of the Hamiltonian therefore uses about four times as many FFTs as in the GGA case. Since the eigensolver applies the Hamiltonian many times in every SCF iteration, this is the leading contribution to the overhead. In DFT-FE, the τ-dependent terms change only the entries of the cell-level Hamiltonian matrices in Eq. (34). Each application of the Hamiltonian therefore costs the same as in the GGA case. The extra work is limited to evaluating τ^σ and the XC derivatives at the quadrature points once per SCF iteration. This needs no global transforms and adds little to the cost of an SCF iteration. The measured costs reflect this difference between the two codes.

Figure 3a, b shows the ratio of the r^{2}SCAN to the PBE cost per SCF iteration in each code. In QE, an r^{2}SCAN iteration costs 5.4 times as much as a PBE iteration for the smallest $BaTiO_3$ supercell. The ratio decreases with system size but is still 1.7 for the largest supercell that QE could run. The Pt nanoparticles show the same behavior in QE, with ratios between 2.5 and 4.8. In DFT-FE, the ratio falls from 1.62 to 1.10 as the $BaTiO_3$ supercell grows, and it stays between 1.12 and 1.15 for all Pt nanoparticles.

The ratios above compare each code with itself. Figure 3c, d instead compares the r^{2}SCAN cost per SCF iteration of DFT-FE and QE directly. For the $4 \times 4 \times 4$ $BaTiO_3$ supercell, the two codes have almost the same cost (0.9×). The cost of DFT-FE grows more slowly with system size, and DFT-FE becomes faster from the $5 \times 5 \times 5$ supercell onward. Its speedup over QE grows with system size and reaches 4.3× for the $7 \times 7 \times 7$ supercell. For the Pt nanoparticles, DFT-FE is faster than QE at every size, with speedups between 1.8× and 4.0×, the largest for the 561-atom nanoparticle.

For the largest system of each set, the $8 \times 8 \times 8$ $BaTiO_3$ supercell and the 923-atom Pt nanoparticle, the QE runs fail during the first SCF iteration, even with PBE. We attribute this to the memory and MPI parallelization limits of QE on the Polaris GPU nodes. DFT-FE completes both systems with r^{2}SCAN on the same hardware, using 64 and 12 Polaris nodes, respectively.

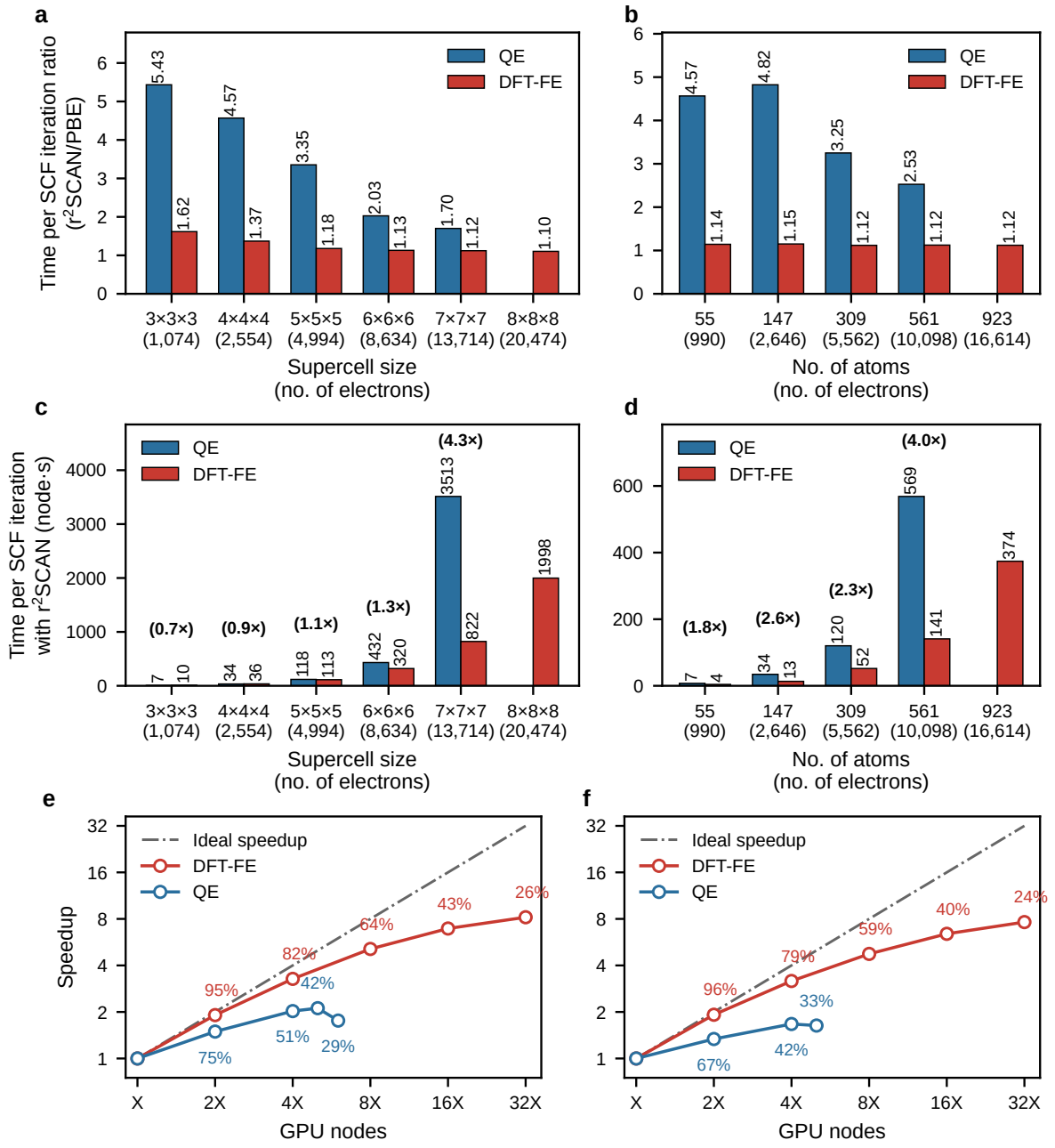


Fig. 3 Performance of DFT-FE and QE with r^{2}SCAN for $BaTiO_3$ supercells (left) and Pt nanoparticles (right) of increasing size. **a**, **b** Ratio of r^{2}SCAN to PBE computational cost per SCF iteration. **c**, **d** Computational cost per SCF iteration with r^{2}SCAN. The bold labels in parentheses denote the speedup of DFT-FE over QE. **e**, **f** Parallel strong scaling of DFT-FE (on Aurora) and QE (on Polaris) for the $7 \times 7 \times 7$ $BaTiO_3$ supercell with 13714 electrons (**e**) and the 561-atom Pt nanoparticle with 10098 electrons (**f**). Percentages denote the scaling efficiency at each node count relative to the baseline X, and the dash-dotted line is the ideal speedup. In all panels, the blue bars and circles are QE and the red bars and circles are DFT-FE.

Scaling comparisons

The parallel design of DFT-FE allows it to keep gaining speed at large node counts. DFT-FE distributes the FE mesh over the MPI processes by domain decomposition. Because the FE basis functions have compact support, the action of the Hamiltonian is computed with cell-level matrices local to each subdomain, followed by nearest-neighbor communication of the values at the nodes shared with neighboring subdomains. The volume of this communication scales with the surface of a subdomain, whereas the computation scales with its volume. The ratio of communication to computation therefore grows only slowly as the subdomains shrink. Every major step of an SCF iteration therefore speeds up as nodes are added. The measured scaling confirms this.

Figure 3e, f shows the parallel strong scaling of DFT-FE and QE for the $7 \times 7 \times 7$ $BaTiO_3$ supercell and the 561-atom Pt nanoparticle. The speedup in wall time per SCF iteration is measured relative to the minimum number of nodes X on which each problem fits in memory, with DFT-FE run on Aurora and QE on Polaris. At $4X$ nodes, DFT-FE retains an efficiency of 82% for $BaTiO_3$ and 79% for Pt. It continues to gain speed up to $32X$ nodes, where it runs 8.2 and 7.6 times faster than at X, respectively. QE loses efficiency faster. Its speedup saturates at $5X$ nodes for $BaTiO_3$ and at $4X$ for Pt, where the efficiency has dropped to 42% in both cases.

Table 2 compares the minimum wall time per SCF iteration of the two codes on Polaris. Since DFT-FE keeps gaining speed at node counts where QE has saturated, it reaches a T_{min} that is 11.1× to 15.2× shorter on the same hardware.

Table 2 Minimum wall time per SCF iteration (T_{min}) with r^{2}SCAN for DFT-FE and QE for the $BaTiO_3$ supercells and the 561-atom Pt nanoparticle, measured on ALCF Polaris (NVIDIA A100 GPUs). For each code, T_{min} is reported with the node count at which it is realized and the corresponding parallel efficiency (Eff.).

		DFT-FE			QE			
System	Electrons	Nodes	T_{min} (s)	Eff.	Nodes	T_{min} (s)	Eff.	Speedup
$BaTiO_3$ $6 \times 6 \times 6$	8634	256	4.5	27%	24	50.6	36%	11.1×
$BaTiO_3$ $7 \times 7 \times 7$	13714	264	7.3	43%	95	87.4	42%	11.9×
Pt_{561}	10098	112	4.5	28%	20	68.0	42%	15.2×

Large-scale demonstrations of accuracy and efficiency

Oxygen vacancy formation energy in $BaTiO_3$

$BaTiO_3$ is a ferroelectric perovskite that changes structure on cooling. It is cubic above 393 K, tetragonal below this temperature, and orthorhombic and rhombohedral at lower temperatures.[50,51] The tetragonal phase is therefore the room-temperature structure and is the reference used here for the vacancy calculations. We first check that r^{2}SCAN reproduces the tetragonal distortion by computing the c/a ratio. Our calculations give $c/a = 1.027$, close to the SCAN value of 1.029 reported by Sun *et al.*[12] and to the experimental value of 1.010. PBE gives 1.054, i.e. too large a distortion, because it overestimates the cell volume.[12]

The tetragonal structure has two inequivalent oxygen sites, an equatorial site in the TiO_2 plane and an axial site in the BaO plane. Using the B1WC hybrid functional, Rusevich *et al.*[50] found the axial vacancy to be the more stable of the two, with a formation energy of 7.32 eV against 7.48 eV. All vacancy calculations reported here therefore use the axial BaO-plane site and vacancies are considered to be neutral.

The formation energy of the neutral vacancy is computed as

$$E_{\mathrm{vac}} = E(V_{\mathrm{O}}) + \frac{1}{2}\,E(\mathrm{O}_2) - E(\mathrm{perfect}), \tag{40}$$

where $E(V_{\mathrm{O}})$ and $E(\mathrm{perfect})$ are the r^{2}SCAN total energies of the defective and perfect supercells and $\frac{1}{2}E(\mathrm{O}_2)$ is the reference energy per oxygen atom, taken from the gas-phase O_2 molecule in its triplet ground state.[50] The primitive cell is first relaxed with r^{2}SCAN, both ionic positions and cell parameters, and the supercells are built from the relaxed cell. One oxygen atom is then removed and the ionic positions are relaxed at fixed cell shape and volume.

Figure 4a shows E_{vac} for supercells from $2 \times 2 \times 2$ (40 atoms) to $8 \times 8 \times 8$ (2560 atoms). The formation energy decreases with supercell size, from 6.18 eV to 5.29 eV, as the interaction between the vacancy and its periodic images is reduced. The two largest supercells differ by 0.03 eV, so we take the $8 \times 8 \times 8$ value of 5.29 eV as the dilute-limit result. The largest supercell contains 2560 atoms and about 20500 valence electrons. Calculations of this size are intractable for plane-wave codes at the τ-mGGA level, as the performance benchmarks above show. They are possible in DFT-FE because a τ-mGGA calculation costs little more than a GGA calculation in those cases.

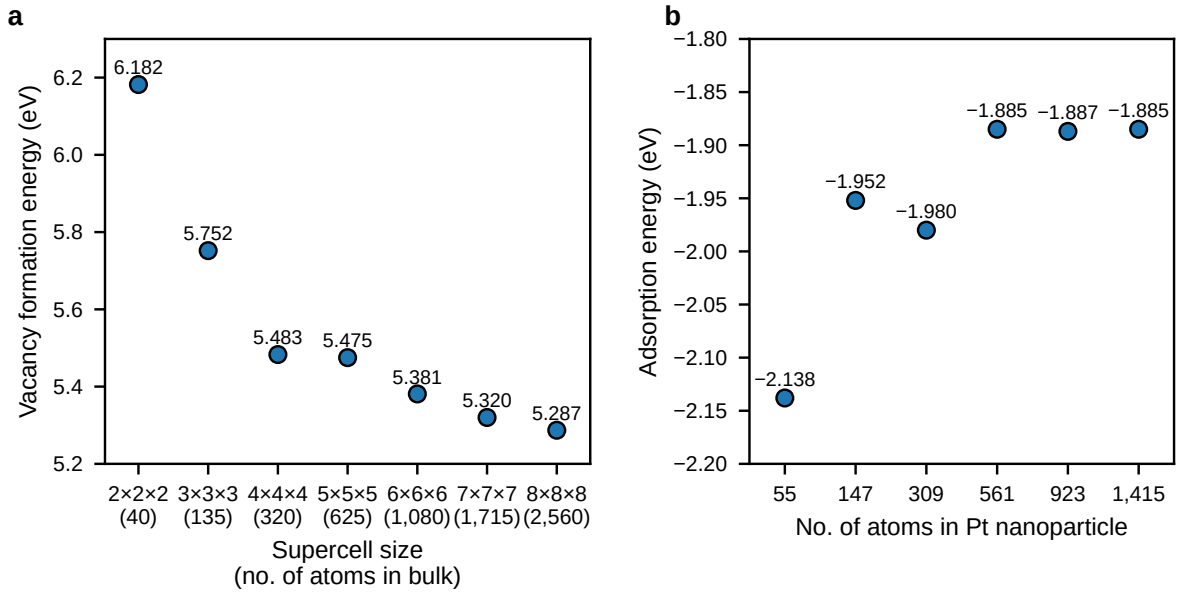


Fig. 4 Convergence of the two application energies with system size, computed with τ-mGGA functionals in DFT-FE. **a** Neutral oxygen vacancy formation energy in tetragonal $BaTiO_3$ with r^{2}SCAN, for supercells from $2 \times 2 \times 2$ (40 atoms) to $8 \times 8 \times 8$ (2560 atoms). **b** CO adsorption energy at the (211)-like step site of cuboctahedral Pt nanoparticles with M06-L, for nanoparticles from 55 to 1415 atoms, the largest corresponding to 25470 valence electrons.

CO adsorption energy at step sites of Pt nanoparticles

CO adsorption on Pt is a standard test case in heterogeneous catalysis and is relevant to fuel-cell poisoning and CO oxidation.[52,53] Pt catalysts are usually nanoparticles a few nanometers in size. On metal particles this small, the adsorption energy changes with particle size, and not always monotonically.[54] DFT studies of finite-size effects in Au and Pt nanoparticles of up to 1415 atoms have determined the particle size at which adsorption energies converge to the extended-surface limit.[55,56] This size differs between metals, which Li *et al.* traced to differences in the electronic structure near the Fermi level.[56] It may therefore also depend on the XC functional. For CO adsorption on Pt, a τ-mGGA is more accurate than a GGA, as discussed below. To our knowledge, finite-size effects on adsorption energies have not been studied at the τ-mGGA level.

Here we determine the converged size for CO adsorption at the step sites of Pt nanoparticles. The largest particle considered for this study consists of 1415 atoms, and it is about 4 nm across. To show that the adsorption energy has converged, the series has to extend past the size at which it stops changing. Plane-wave codes cannot simulate such large particles and the vacuum around them at the τ-mGGA level, as the performance benchmarks show.

No GGA describes CO adsorption energies and metal surface energies well at the same time.[57,58] PBE overestimates CO adsorption energies and places CO in the wrong site on Pt(111).[59,60] RPBE, which was designed for chemisorption, gives the smallest error in CO adsorption energies of the functionals that Luo *et al.* tested on five transition metals, but it underestimates surface formation energies by 0.37 eV on average and places CO in the hollow site on Pt(111).[58] M06-L[61] removes much of this imbalance at semilocal cost. Across the same five metals, its mean unsigned errors are 0.17 eV for CO adsorption energies and 0.07 eV for surface formation energies, and it predicts the experimental adsorption site for four of the five metals, including the on-top site on Pt(111).[58] We therefore use M06-L for the adsorption energy calculations reported here.

On a Pt catalyst particle, the atoms at steps, edges, and corners have fewer neighbors than those in the interior of a facet, and CO binds more strongly at these sites.[62] We compute the adsorption energy at the Pt(211)-like step site of cuboctahedral Pt nanoparticles, that is, the on-top site of a seven-fold coordinated edge atom, which has the same number of Pt neighbors as the step atom of the Pt(211) surface. At low coverage, LEED and electron-stimulated desorption measurements place CO at the one-fold step-edge site on Pt(211), which is the site considered here.[63] We use the closed-shell cuboctahedral sizes of 55, 147, 309, 561, 923, and 1415 atoms. The nanoparticles are constructed at the bulk fcc lattice constant optimized with M06-L. The Pt atoms are then held fixed, and only the C and O atoms are

relaxed in the adsorption calculations. Increasing the particle size reduces the effect of the finite facet size. The size-converged value for the largest particle therefore approximates the adsorption energy at an extended step. The adsorption energy is calculated as

$$E_{\mathrm{ads}} = E(\mathrm{CO/Pt}_N) - E(\mathrm{Pt}_N) - E(\mathrm{CO}), \tag{41}$$

where $E(\mathrm{CO/Pt}_N)$ and $E(\mathrm{Pt}_N)$ are the total energies of the nanoparticle with and without the adsorbed CO molecule and $E(\mathrm{CO})$ is the energy of the isolated CO molecule. All calculations use Γ-point sampling, as the nanoparticles are non-periodic.

Figure 4b shows the adsorption energy as a function of particle size. The 55-atom nanoparticle binds CO more strongly (−2.138 eV) than any of the larger particles. From 147 atoms onward the values lie within 0.10 eV of each other, being −1.952, −1.980, −1.885, −1.887, and −1.885 eV for the 147, 309, 561, 923, and 1415 atom nanoparticles. The three largest particles agree to within 0.002 eV, so the adsorption energy at the step site has converged with particle size at 561 atoms. We take the 1415-atom value, −1.885 eV, as the M06-L adsorption energy at the step site. Single-crystal adsorption calorimetry on a Pt(211) surface gives an adsorption energy of −1.917 eV per CO molecule in the zero-coverage limit,[64] which this value reproduces to within 0.03 eV.

Flat bands in twisted bilayer MoS_2

Two layers of a van der Waals material stacked at a small twist angle θ form a moiré superlattice. As θ decreases, the bands near the band edges flatten. Flat bands enhance the effect of electron interactions and can give rise to correlated phases.[65] In twisted bilayer MoS_2, Xian *et al.*[65] found two families of flat valence bands that derive from the Γ point of the untwisted bilayer. The first family resembles the bands of graphene, with a Dirac point at K. The second realizes a p_x–p_y honeycomb model, in which destructive interference produces nearly dispersionless bands. These bands appear only at small twist angles, where the moiré cell is large. A commensurate bilayer with integer index m has a twist angle and a number of atoms given by[66]

$$\cos\theta = \frac{3m^2 + 3m + 1/2}{3m^2 + 3m + 1} \quad \text{and} \quad N_{\mathrm{at}} = 6\left(3m^2 + 3m + 1\right). \tag{42}$$

Twist angles near 2° therefore need cells with several thousand atoms. Xian *et al.* relaxed these cells and computed their bands with the PBE functional[7] and the Tkatchenko-Scheffler (TS) dispersion correction. We repeat this study at the τ-mGGA level, for both the structures and the bands, with the same cells $m = 10$, 12 and 14 ($\theta = 3.15°$, 2.65° and 2.28°).

For the bands, we use the LAK τ-mGGA functional proposed by Lebeda, Aschebrock and Kümmel.[67] It exploits a freedom in the gradient expansion that earlier meta-GGAs did not use, and this lets a single functional describe both band gaps and binding energies.[67,68] For fifteen semiconductors, its band gaps have a mean absolute error of 0.18 eV, close to the 0.17 eV of the HSE06 hybrid functional,[69,70] although LAK contains no exact exchange.[67,68] LAK thus offers band gaps of hybrid quality at the cost of a semilocal functional. In DFT-FE, a τ-mGGA costs little more than a GGA, as the performance benchmarks above show. This makes LAK well suited to moiré cells with thousands of atoms.

For the structures, LAK alone is not sufficient. Without a dispersion correction, it overestimates the in-plane lattice constant of bulk 2H-MoS_2 by 1.6% and the interlayer distance by 5.1%. The TS correction cannot be added to LAK in the usual way. Its damping parameter s_R is normally fitted to the S22 interaction energies.[71–73] LAK, however, already describes weak interactions near equilibrium without a dispersion correction.[74] On its own, it overbinds 18 of the 22 S22 complexes, with a mean absolute error of 0.43 kcal/mol, and adding TS increases this error for every value of s_R. S22 therefore gives no finite s_R for LAK. A fit to binding energies of transition-metal dichalcogenide bilayers from the random phase approximation (RPA)[75] gives $s_R = 1.73$. With this value, LAK+TS still overestimates the in-plane lattice constant by 1.5% and the interlayer distance by 4.8%, almost as much as LAK alone.

We therefore relax the atomic positions with r^{2}SCAN and the TS correction,[15,71] which is included self-consistently.[76] For r^{2}SCAN, the S22 fit gives a well-defined damping parameter, $s_R = 1.165$, with a mean absolute error of 0.27 kcal/mol. With this value, r^{2}SCAN+TS reproduces the measured lattice constants a and c of bulk 2H-MoS_2 to within 0.1% and 0.4%, respectively.[77] The LAK bands are then computed on these structures without TS. Computing the bands at a structure obtained with another

method is common practice. Borlido et al. compared twelve functionals on the band gaps of 472 solids, all at the experimental crystal structures.[78] He et al. obtained the interlayer distances of transition-metal dichalcogenide bilayers from RPA and computed the electronic and optical properties at these distances with the GW approximation and the Bethe–Salpeter equation.[75] Neither study relaxes the structure with the method used for the electronic structure.

We test this combination on bulk 2H-MoS_2, an indirect-gap semiconductor with a measured gap of 1.29 eV.[79] Its valence-band maximum (VBM) lies at Γ. Its conduction-band minimum (CBM) lies in the Λ valley, about halfway between Γ and K.[79,80] Table 3 lists the computed gaps. At the r^{2}SCAN+TS relaxed structure, LAK gives an indirect gap of 1.03 eV, against 0.95 eV for r^{2}SCAN+TS. Both functionals place the band edges where they are observed, with the VBM at Γ and the CBM at Λ. In LAK, the Λ valley lies 0.08 eV below the conduction band at K. LAK still underestimates the gap by 0.26 eV. It is smaller than the underestimate of 0.44 eV for PBE. At the experimental structure, LAK gives the same gap of 1.03 eV, at the same location. The r^{2}SCAN+TS relaxed structure is therefore as good as the measured one for the band edges. At the LAK structure, where the layers are too far apart, the CBM moves to K.

Table 3 Indirect band gap E_g of bulk 2H-MoS_2 and the location of its CBM for different structures and functionals. The same functional is used for the SCF and the bands. The VBM is at Γ in all cases. The experimental structure has $a = 3.160$ Å and $c = 12.29$ Å.[77]

Structure	Functional	E_g (eV)	CBM
r^{2}SCAN+TS relaxed	LAK	1.029	Λ
r^{2}SCAN+TS relaxed	r^{2}SCAN+TS	0.953	Λ
Experimental	LAK	1.032	Λ
Experimental	r^{2}SCAN	0.967	Λ
Experimental	PBE	0.854	Λ
LAK relaxed	LAK	1.085	K
Measured[79,80]		1.29	Λ

With the recipe tested on bulk MoS_2, we turn to the twisted bilayers. Their cells are built with the in-plane lattice constant of bulk 2H-MoS_2 from r^{2}SCAN+TS, $a = 3.163$ Å. They are periodic in the plane of the layers and non-periodic along the normal, with at least 25 bohr of vacuum on each side. We use FE polynomial degree $p = 7$ with elements of 1.6 bohr near the atoms, and Γ-point sampling in the SCF iterations. In the relaxed structures, the largest force component is below 5×10^{-4} Ha/bohr and the in-plane cell stress is below 5×10^{-6} Ha/bohr^3. The bands are then computed at k-points along the path Γ–K–M–Γ of the moiré Brillouin zone. The bands of interest are the flat bands at the top of the valence band. They derive from the valence states of MoS_2 at Γ, which are built from out-of-plane Mo d_{z^2} and S p_z orbitals.[65] Spin-orbit coupling splits the valence band of monolayer MoS_2 at K by about 0.16 eV,[81,82] but in twisted bilayer MoS_2 at 3.15° it leaves the bands within 0.4 eV of the VBM unchanged.[65] Therefore spin-orbit coupling is not included.

After relaxation, the interlayer distance varies across the moiré cell. The vertical Mo–Mo distance between the layers is 6.70 Å in the AA regions and 6.14–6.15 Å in the AB and BA regions, in all three cells. These values agree to within 0.04 Å with the r^{2}SCAN+TS interlayer distances of bulk MoS_2 in AA and AB (3R) stacking. The three cells therefore differ in the size of the stacking regions, not in their local structure.

Table 4 Twisted bilayer MoS_2 cells and their LAK band edges and flat-band widths. E_g is the indirect gap from Γ to K of the moiré Brillouin zone. W_1 and W_2 are the widths of the first and second band families, Δ_{12} is the gap between them, and W_{VB3} is the width of the ultra-flat band at the top of the second family.

θ	m	Atoms	Electrons	E_g (eV)	W_1 (meV)	Δ_{12} (meV)	W_2 (meV)	W_{VB3} (meV)
3.15°	10	1986	17212	1.415	33.2	22.4	86.1	3.8
2.65°	12	2814	24388	1.410	20.0	24.5	59.0	1.4
2.28°	14	3786	32812	1.406	12.4	24.9	42.7	0.6

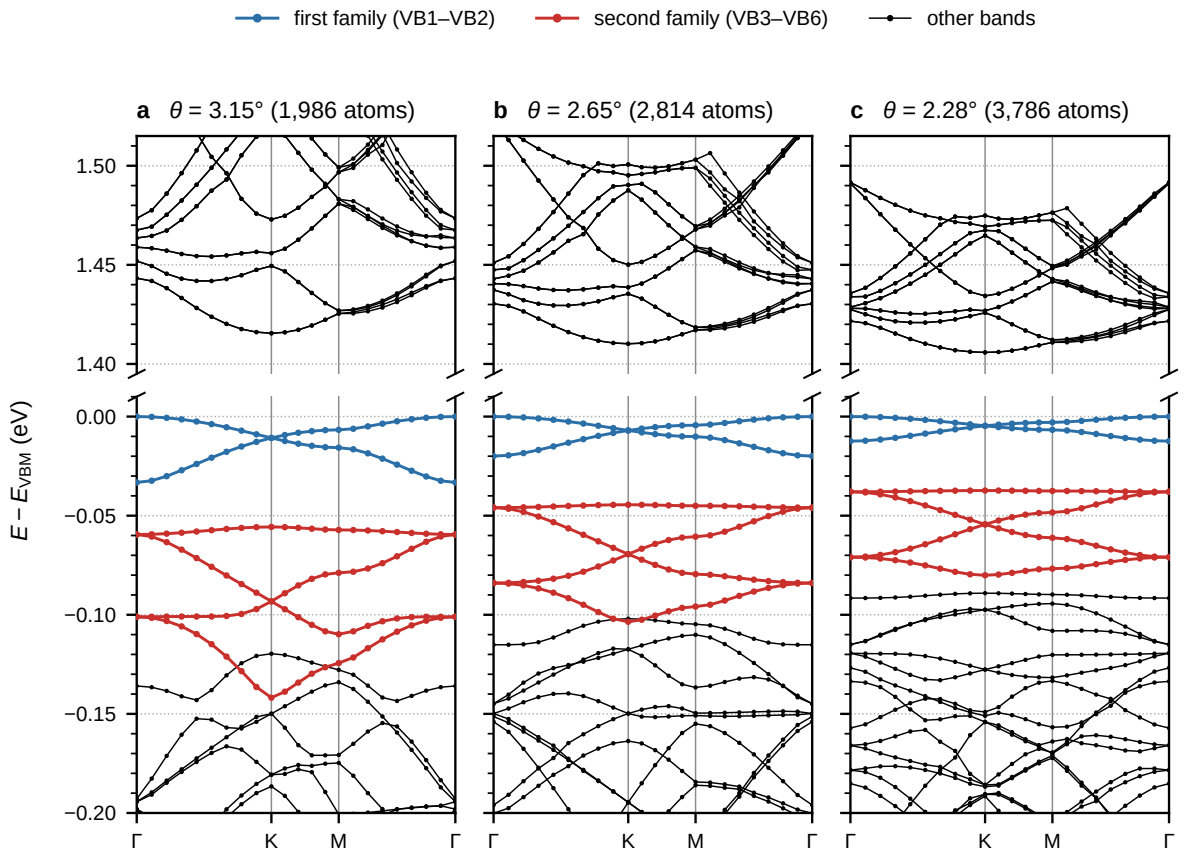


Fig. 5 Band edges of twisted bilayer MoS_2 at three twist angles, computed with LAK on the r^{2}SCAN+TS relaxed structures in DFT-FE. **a** $\theta = 3.15°$. **b** $\theta = 2.65°$. **c** $\theta = 2.28°$. In each panel, the lowest conduction bands are shown above the break in the energy axis and the top valence bands below it. Energies are relative to the VBM. Bands are numbered at Γ from the VBM (VB1). The first family of flat bands is shown in blue and the second in red. Each dot is an eigenvalue at a computed k-point.

Figure 5 shows the band edges of the three cells. The valence bands follow the layout of Fig. 1b of Xian *et al.*[65] Table 4 lists the gaps and the bandwidths. In all three cells the VBM is at Γ and the CBM at the K point of the moiré Brillouin zone. A VBM at Γ agrees with angle-resolved photoemission (ARPES) measurements on 2H-stacked[83] and twisted[84] bilayer MoS_2, irrespective of the twist angle. The gap decreases by only 10 meV between 3.15° and 2.28°. The lowest conduction bands come in nearly degenerate pairs. The lowest pair narrows from 27.8 meV at 3.15° to 15.8 meV at 2.28°, more slowly than the valence bands. As found by Xian *et al.*, two families of bands separate from the other bands as θ decreases. The first family (VB1–VB2) has two bands that touch at K and form a Dirac point. Its width falls from 33.2 meV at 3.15° to 12.4 meV at 2.28°. The second family (VB3–VB6) has four bands. VB3 and VB4 touch at Γ, VB4 and VB5 cross at K, and VB5 and VB6 touch at Γ. The top band of this family, VB3, is ultra-flat, and its width falls from 3.8 meV to 0.6 meV. The widths of both families are close to those reported by Xian *et al.*

Two features differ from the PBE+TS results. First, the edge bands of the second family, VB3 and VB6, are flatter at small angles than in Fig. 1b of Xian *et al.* Second, at 2.28° the second family is fully separated from the lower bands, and a further flat band, 2.5 meV wide, lies 9.0 meV below it. Xian *et al.* expected the next family of flat bands at smaller angles. To separate the effects of the structure and of the band functional, we also computed the bands of the 3.15° cell on a structure relaxed with PBE+TS in DFT-FE. On this structure, LAK widens each of VB3, VB6 and VB7 by 1 to 7 meV compared with PBE+TS. Going from the PBE+TS relaxed structure to the r^{2}SCAN+TS relaxed structure narrows them by up to 2.5 meV. LAK therefore does not flatten the bands. The flatter bands and the further flat band at small angles most likely come from the r^{2}SCAN+TS relaxed structures.

Discussion

We have presented a real space finite-element methodology for large-scale KS-DFT calculations with τ-mGGA functionals, implemented within the open-source DFT-FE framework. Atomic forces and periodic unit cell stresses are computed via a unified generalized force approach extended to the τ-mGGA setting. The accuracy of the implementation was validated against Quantum ESPRESSO, with ground-state energies agreeing well within chemical accuracy and the consistency of forces and stresses confirmed via energy-derivative benchmarks.

The central result of this work is that the proposed methodology achieves τ-mGGA computational costs close to those of GGA in the real space FE framework. The ratio of r^{2}SCAN to PBE cost per SCF iteration in DFT-FE decreases with increasing size of the periodic $BaTiO_3$ supercells and stays near unity for the Pt nanoparticles, demonstrating that the τ-mGGA overhead is small. This stands in sharp contrast to the plane-wave setting, where the τ-mGGA overhead remains substantial due to the additional FFT evaluations and finer auxiliary grids required to handle τ. The small overhead in DFT-FE is a direct consequence of the real space formulation, where τ and its functional derivatives are evaluated locally at the quadrature points using the same infrastructure as the density and its gradient, incurring no additional global communication or transformation cost.

In absolute terms, DFT-FE with r^{2}SCAN achieves significant speedups over QE for both periodic and non-periodic systems, with the gains increasing with system size. For the largest systems considered, QE could not accommodate the calculations on the available GPU hardware, while DFT-FE completes them at moderate cost. This advantage stems from the locality and compact support of the FE basis functions, which yield sparse Hamiltonian matrices amenable to efficient parallel distribution, in contrast to the dense all-to-all communication patterns of global FFTs in plane-wave codes. Strong scaling studies confirm this advantage, with DFT-FE sustaining meaningful speedups well beyond the point where QE saturates, for both periodic and non-periodic systems. Because DFT-FE continues to benefit from added resources at concurrencies where QE no longer does, it also attains a much shorter minimum wall time per SCF iteration on identical hardware. On ALCF Polaris, the minimum wall times of DFT-FE and QE are 4.5 s and 50.6 s for the $6\times6\times6$ $BaTiO_3$ supercell, 7.3 s and 87.4 s for $7\times7\times7$ $BaTiO_3$ supercell, and 4.5 s and 68.0 s for the 561-atom Pt nanoparticle, corresponding to 11.1×, 11.9× and 15.2× reductions in time per SCF. This is of direct consequence for *ab initio* molecular dynamics and geometry optimization, where a long sequence of SCF solves must be performed one after another.

Near-GGA cost at the τ-mGGA level also matters for the reference data used to train machine-learned interatomic potentials. These models, from neural-network and kernel-based constructions [85,86] to universal foundation potentials [87], are limited in accuracy by the level of theory of their training labels. Yet the largest public datasets are mostly at the GGA level. OMat24, for instance, contains more than 110 million PBE and PBE+U calculations [88], and models trained on such data inherit the systematic errors of the underlying functional. This has motivated τ-mGGA reference datasets, such as the r^{2}SCAN releases of MatPES [89] and MP-ALOE [90], as well as schemes that combine energies computed at different levels of theory [19]. These datasets, however, are built almost entirely from cells of a few tens of atoms, because plane-wave τ-mGGA calculations are expensive. DFT-FE provides τ-mGGA energies, forces and stresses at near-GGA cost, at a few seconds per SCF iteration for ten thousand electrons. This makes τ-mGGA labels affordable for the extended defects, nanoparticles, surfaces and interfaces that are underrepresented in existing datasets. The unit cell stresses also allow training configurations to be sampled from variable-cell molecular dynamics, rather than from fixed-cell snapshots alone.

The practical significance of achieving near-GGA cost at the τ-mGGA level was demonstrated through three large-scale applications. The neutral oxygen vacancy formation energy in a ferroelectric perovskite was converged to the dilute limit using supercells containing over 2500 atoms, and the CO adsorption energy at the (211)-like step site of Pt nanoparticles was converged with respect to particle size using nanoparticles of up to 1415 atoms and agrees with the calorimetric value to within 0.03 eV. The band structure of twisted bilayer MoS_2, with its flat bands, was analyzed in moiré cells of up to 3786 atoms. These calculations, involving up to 32812 valence electrons at the τ-mGGA level, are well beyond the reach of plane-wave codes, underscoring the potential of the proposed methodology to enable *ab initio* materials modeling at length scales and levels of XC accuracy that were previously inaccessible.

Methods

Pseudopotentials. All DFT-FE and QE calculations in this work use the SG15 set[91] of ONCV pseudopotentials,[43] which are generated with PBE.

Accuracy benchmarks. For the comparison of r^{2}SCAN ground-state free energies with QE, QE uses a kinetic energy cutoff `ecutwfc` = 200 Ry, and DFT-FE uses a 7th order FE interpolating polynomial with a mesh size of 0.8 Bohr. With these settings, the energies are converged well below the differences reported in Table 1.

Performance benchmarks. The periodic set consists of $BaTiO_3$ supercells, each containing a single oxygen vacancy, ranging from $3 \times 3 \times 3$ (1074 electrons) to $8 \times 8 \times 8$ (20474 electrons). The non-periodic set consists of cuboctahedral Pt nanoparticles ranging from 55 atoms (990 electrons) to 923 atoms (16614 electrons). In DFT-FE, both sets use FE basis functions of degree $p = 7$, with a mesh size of 1.25 Bohr for $BaTiO_3$ and 1.65 Bohr for the Pt nanoparticles. The quadrature rule has 14 points per direction for PBE and 16 for r^{2}SCAN in $BaTiO_3$, and 10 points per direction for both functionals in the Pt nanoparticles. In QE, the $BaTiO_3$ calculations use `ecutwfc` = 90 Ry, with `ecutrho` = 540 Ry for PBE and 720 Ry for r^{2}SCAN. The Pt calculations use `ecutwfc` = 65 Ry and `ecutrho` = 260 Ry for both functionals. These parameters give a discretization error in the ground-state energy within $\mathcal{O}(10^{-4})$ Ha/atom in both codes. All calculations in both series use Γ-point sampling.

All comparisons are carried out on GPU architectures using three metrics. The first is the computational cost per SCF iteration in node·seconds (node·s), defined as the product of the minimum number of nodes on which the problem fits in memory and the corresponding average wall time per SCF iteration. The second is the parallel strong scaling efficiency, obtained from the speedup in wall time per SCF iteration as the number of nodes is increased from this minimum. The third is the minimum wall time per SCF iteration, T_{min}, which is the lowest wall time per iteration that each code attains. In all reported timings, the first SCF iteration is excluded from the per-iteration cost, as it is typically more expensive than subsequent iterations and its weight in the average depends on the total number of SCF iterations. The node·s and T_{min} comparisons between DFT-FE and QE are performed on Polaris. Polaris is a GPU-accelerated supercomputer at the Argonne Leadership Computing Facility (ALCF), comprising 560 nodes, each equipped with one AMD EPYC Milan 7543P processor (32 cores) and four NVIDIA A100 GPUs. The parallel scaling studies for DFT-FE are carried out on Aurora to demonstrate the scalability of our implementation on a leadership-class exascale architecture. Aurora is a GPU-accelerated exascale supercomputer at the ALCF, comprising 10,624 nodes, each equipped with two Intel Xeon CPU Max Series processors and six Intel Data Center GPU Max Series (Ponte Vecchio) GPUs. DFT-FE runs on the Intel GPUs of Aurora through its SYCL backend. QE 7.5, the version used here, has no SYCL backend, and its GPU build supports only NVIDIA GPUs. The QE scaling studies are therefore performed on Polaris.

References

[1] Hohenberg, P. & Kohn, W. Inhomogeneous Electron Gas. *Phys. Rev.* **136,** B864–B871 (1964).

[2] Kohn, W. & Sham, L. J. Self-Consistent Equations Including Exchange and Correlation Effects. *Phys. Rev.* **140,** A1133–A1138 (1965).

[3] Jones, R. O. Density functional theory: Its origins, rise to prominence, and future. *Rev. Mod. Phys.* **87,** 897–923 (2015).

[4] Martin, R. M. *Electronic structure: basic theory and practical methods* (Cambridge university press, 2020).

[5] Perdew, J. P. & Schmidt, K. Jacob's ladder of density functional approximations for the exchange-correlation energy. *AIP Conf. Proc.* **577,** 1–20 (2001).

[6] Perdew, J. P. & Wang, Y. Accurate and simple analytic representation of the electron-gas correlation energy. *Phys. Rev. B* **45,** 13244–13249 (1992).

[7] Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **77,** 3865–3868 (1996).

[8] Csonka, G. I. et al. Assessing the performance of recent density functionals for bulk solids. *Phys. Rev. B* **79,** 155107 (2009).

[9] Isaacs, E. B. & Wolverton, C. Performance of the strongly constrained and appropriately normed density functional for solid-state materials. *Phys. Rev. Mater.* **2,** 063801 (2018).

[10] Kaplan, A. D., Levy, M. & Perdew, J. P. The Predictive Power of Exact Constraints and Appropriate Norms in Density Functional Theory. *Annu. Rev. Phys. Chem.* **74,** 193–218 (2023).

[11] Sun, J., Ruzsinszky, A. & Perdew, J. P. Strongly Constrained and Appropriately Normed Semilocal Density Functional. *Phys. Rev. Lett.* **115,** 036402 (2015).

[12] Sun, J. et al. Accurate first-principles structures and energies of diversely bonded systems from an efficient density functional. *Nat. Chem.* **8,** 831–836 (2016).

[13] Bartók, A. P. & Yates, J. R. Regularized SCAN functional. *J. Chem. Phys.* **150,** 161101 (2019).

[14] Lehtola, S. & Marques, M. A. L. Many recent density functionals are numerically ill-behaved. *J. Chem. Phys.* **157,** 174114 (2022).

[15] Furness, J. W., Kaplan, A. D., Ning, J., Perdew, J. P. & Sun, J. Accurate and Numerically Efficient r2SCAN Meta-Generalized Gradient Approximation. *J. Phys. Chem. Lett.* **11,** 8208–8215 (2020).

[16] Kingsbury, R. et al. Performance comparison of r^2SCAN and SCAN metaGGA density functionals for solid materials via an automated, high-throughput computational workflow. *Phys. Rev. Mater.* **6,** 013801 (2022).

[17] Kothakonda, M. et al. Testing the r2SCAN Density Functional for the Thermodynamic Stability of Solids with and without a van der Waals Correction. *ACS Mater. Au* **3,** 102–111 (2022).

[18] Rezaei, N., Alaei, M. & Oganov, A. R. Evaluating SCAN and r^{2}SCAN meta-GGA functionals for predicting transition temperatures in antiferromagnetic materials. *Phys. Rev. B* **111,** 144406 (2025).

[19] Kingsbury, R. S. et al. A flexible and scalable scheme for mixing computed formation energies from different levels of theory. *npj Comput. Mater.* **8,** 195 (2022).

[20] Womack, J. C., Mardirossian, N., Head-Gordon, M. & Skylaris, C.-K. Self-consistent implementation of meta-GGA functionals for the ONETEP linear-scaling electronic structure package. *J. Chem. Phys.* **145,** 204114 (2016).

[21] Doumont, J., Tran, F. & Blaha, P. Implementation of self-consistent MGGA functionals in augmented plane wave based methods. *Phys. Rev. B* **105,** 195138 (2022).

[22] Yao, Y. & Kanai, Y. Plane-wave pseudopotential implementation and performance of SCAN meta-GGA exchange-correlation functional for extended systems. *J. Chem. Phys.* **146,** 224105 (2017).

[23] Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54,** 11169–11186 (1996).

[24] Giannozzi, P. et al. Advanced capabilities for materials modelling with Quantum ESPRESSO. *J. Phys. Condens. Matter* **29,** 465901 (2017).

[25] Carnimeo, I. et al. Quantum ESPRESSO: One Further Step toward the Exascale. *J. Chem. Theory Comput.* **19,** 6992–7006 (2023).

[26] Soler, J. M. et al. The SIESTA method for ab initio order-N materials simulation. *J. Phys. Condens. Matter* **14,** 2745 (2002).

[27] Skylaris, C.-K., Haynes, P. D., Mostofi, A. A. & Payne, M. C. Introducing ONETEP: Linear-scaling density functional simulations on parallel computers. *J. Chem. Phys.* **122,** 084119 (2005).

[28] Liu, R. et al. Implementation of the meta-GGA exchange-correlation functional in numerical atomic orbital basis: With systematic testing on SCAN, rSCAN, and r2SCAN functionals. *J. Chem. Phys.* **159,** 074109 (2023).

[29] Chelikowsky, J. R., Troullier, N. & Saad, Y. Finite-difference-pseudopotential method: Electronic structure calculations without a basis. *Phys. Rev. Lett.* **72,** 1240–1243 (1994).

[30] Genovese, L. et al. Daubechies wavelets as a basis set for density functional pseudopotential calculations. *J. Chem. Phys.* **129,** 014109 (2008).

[31] White, S. R., Wilkins, J. W. & Teter, M. P. Finite-element method for electronic structure. *Phys. Rev. B* **39,** 5819–5833 (1989).

[32] Tsuchida, E. & Tsukada, M. Electronic-structure calculations based on the finite-element method. *Phys. Rev. B* **52,** 5573–5578 (1995).

[33] Tsuchida, E. & Tsukada, M. Adaptive finite-element method for electronic-structure calculations. *Phys. Rev. B* **54,** 7602–7605 (1996).

[34] Motamarri, P., Nowak, M., Leiter, K., Knap, J. & Gavini, V. Higher-order adaptive finite-element methods for Kohn–Sham density functional theory. *J. Comput. Phys.* **253,** 308–343 (2013).

[35] Motamarri, P. et al. DFT-FE - A massively parallel adaptive finite-element code for large-scale density functional theory calculations. *Comput. Phys. Commun.* **246,** 106853 (2020).

[36] Das, S., Motamarri, P., Subramanian, V., Rogers, D. M. & Gavini, V. DFT-FE 1.0: A massively parallel hybrid CPU-GPU density functional theory code using finite-element discretization. *Comput. Phys. Commun.* **280,** 108473 (2022).

[37] Das, S. et al. Large-Scale Materials Modeling at Quantum Accuracy: Ab Initio Simulations of Quasicrystals and Interacting Extended Defects in Metallic Alloys. in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC '23 (Association for Computing Machinery, New York, NY, USA, 2023). URL https://doi.org/10.1145/3581784.3627037.

[38] Das, S. et al. Fast, scalable and accurate finite-element based ab initio calculations using mixed precision computing: 46 PFLOPS simulation of a metallic dislocation system. in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC '19 (Association for Computing Machinery, New York, NY, USA, 2019). URL https://doi.org/10.1145/3295500.3357157.

[39] Zhou, Y., Chelikowsky, J. R. & Saad, Y. Chebyshev-filtered subspace iteration method free of sparse diagonalization for solving the Kohn–Sham equation. *J. Comput. Phys.* **274,** 770–782 (2014).

[40] Kodali, N., Ramakrishnan, K. & Motamarri, P. Residual-based Chebyshev filtered subspace iteration for Hermitian eigenvalue problems tolerant to inexact matrix-vector products. *Comput. Phys. Commun.* **327,** 110239 (2026).

[41] Das, S. & Gavini, V. Accelerating self-consistent field iterations in Kohn-Sham density functional theory using a low-rank approximation of the dielectric matrix. *Phys. Rev. B* **107,** 125133 (2023).

[42] Ramakrishnan, K., Das, S. & Motamarri, P. Fast and scalable finite-element based approach for density functional theory calculations using projector augmented wave method. *Phys. Rev. B* **111,** 035101 (2025).

[43] Hamann, D. R. Optimized norm-conserving Vanderbilt pseudopotentials. *Phys. Rev. B* **88,** 085117 (2013).

[44] Motamarri, P. & Gavini, V. Configurational forces in electronic structure calculations using Kohn-Sham density functional theory. *Phys. Rev. B* **97,** 165132 (2018).

[45] Rufus, N. D. & Gavini, V. Ionic forces and stress tensor in all-electron density functional theory calculations using an enriched finite-element basis. *Phys. Rev. B* **106,** 085108 (2022).

[46] Kodali, N. & Motamarri, P. Finite-element methods for noncollinear magnetism and spin-orbit coupling in real-space pseudopotential density functional theory. *Phys. Rev. B* **111,** 195129 (2025).

[47] Ramakrishnan, K. & Motamarri, P. Accelerating finite-element-based projector augmented-wave DFT calculations with scalable GPU-centric computational methods. Preprint at https://arxiv.org/abs/2604.26037 (2026).

[48] Eshelby, J. D. The force on an elastic singularity. *Phil. Trans. R. Soc. Lond. A* **244,** 87–112 (1951).

[49] van der Vorst, H. A. A Generalized Lanczos Scheme. *Math. Comput.* **39,** 559–561 (1982).

[50] Rusevich, L. L., Kotomin, E. A., Zvejnieks, G. & Popov, A. I. Ab initio calculations of structural, electronic and vibrational properties of BaTiO3 and SrTiO3 perovskite crystals with oxygen vacancies. *Low Temp. Phys.* **46,** 1185–1195 (2020).

[51] Erhart, P. & Albe, K. Thermodynamics of mono- and di-vacancies in barium titanate. *J. Appl. Phys.* **102,** 084111 (2007).

[52] Chung, D. Y. et al. Inhibition of CO poisoning on Pt catalyst coupled with the reduction of toxic hexavalent chromium in a dual-functional fuel cell. *Sci. Rep.* **4,** 7450 (2014).

[53] Janthon, P., Viñes, F., Sirijaraensre, J., Limtrakul, J. & Illas, F. Adding Pieces to the CO/Pt(111) Puzzle: The Role of Dispersion. *J. Phys. Chem. C* **121,** 3970–3977 (2017).

[54] Yudanov, I. V., Genest, A., Schauermann, S., Freund, H.-J. & Rösch, N. Size Dependence of the Adsorption Energy of CO on Metal Nanoparticles: A DFT Search for the Minimum Value. *Nano Lett.* **12,** 2134–2139 (2012).

[55] Kleis, J. et al. Finite Size Effects in Chemical Bonding: From Small Clusters to Solids. *Catal. Lett.* **141,** 1067–1071 (2011).

[56] Li, L. et al. Investigation of Catalytic Finite-Size-Effects of Platinum Metal Clusters. *J. Phys. Chem. Lett.* **4,** 222–226 (2013).

[57] Schimka, L. et al. Accurate surface and adsorption energies from many-body perturbation theory. *Nat. Mater.* **9,** 741–744 (2010).

[58] Luo, S., Zhao, Y. & Truhlar, D. G. Improved CO Adsorption Energies, Site Preferences, and Surface Formation Energies from a Meta-Generalized Gradient Approximation Exchange–Correlation Functional, M06-L. *J. Phys. Chem. Lett.* **3,** 2975–2979 (2012).

[59] Feibelman, P. J. et al. The CO/Pt(111) Puzzle. *J. Phys. Chem. B* **105,** 4018–4025 (2000).

[60] Patra, A., Peng, H., Sun, J. & Perdew, J. P. Rethinking CO adsorption on transition-metal surfaces: Effect of density-driven self-interaction errors. *Phys. Rev. B* **100,** 035442 (2019).

[61] Zhao, Y. & Truhlar, D. G. A new local density functional for main-group thermochemistry, transition metal bonding, thermochemical kinetics, and noncovalent interactions. *J. Chem. Phys.* **125,** 194101 (2006).

[62] Hammer, B., Nielsen, O. H. & Nørskov, J. K. Structure sensitivity in adsorption: CO interaction with stepped and reconstructed Pt surfaces. *Catal. Lett.* **46,** 31–35 (1997).

[63] Henderson, M. A., Szabó, A. & Yates, J., J. T. The structure of CO on the Pt(112) stepped surface—a sensitive view of bonding configurations using electron stimulated desorption. *J. Chem. Phys.* **91,** 7245–7254 (1989).

[64] Karmazyn, A., Fiorin, V., Jenkins, S. & King, D. First-principles theory and microcalorimetry of CO adsorption on the {211} surfaces of Pt and Ni. *Surf. Sci.* **538,** 171–183 (2003).

[65] Xian, L. et al. Realization of nearly dispersionless bands with strong orbital anisotropy from destructive interference in twisted bilayer MoS2. *Nat. Commun.* **12,** 5644 (2021).

[66] Lopes dos Santos, J. M. B., Peres, N. M. R. & Castro Neto, A. H. Graphene Bilayer with a Twist: Electronic Structure. *Phys. Rev. Lett.* **99,** 256802 (2007).

[67] Lebeda, T., Aschebrock, T. & Kümmel, S. Balancing the Contributions to the Gradient Expansion: Accurate Binding and Band Gaps with a Nonempirical Meta-GGA. *Phys. Rev. Lett.* **133,** 136402 (2024).

[68] Scuseria, G. E. Perspective: The HSE Screened Hybrid and the Band Gap Problem: Origins, Impact, and the Contenders. Preprint at https://arxiv.org/abs/2609.23225 (2026).

[69] Heyd, J., Scuseria, G. E. & Ernzerhof, M. Hybrid functionals based on a screened Coulomb potential. *J. Chem. Phys.* **118,** 8207–8215 (2003).

[70] Krukau, A. V., Vydrov, O. A., Izmaylov, A. F. & Scuseria, G. E. Influence of the exchange screening parameter on the performance of screened hybrid functionals. *J. Chem. Phys.* **125,** 224106 (2006).

[71] Tkatchenko, A. & Scheffler, M. Accurate Molecular Van Der Waals Interactions from Ground-State Electron Density and Free-Atom Reference Data. *Phys. Rev. Lett.* **102,** 073005 (2009).

[72] Jurečka, P., Šponer, J., Černý, J. & Hobza, P. Benchmark database of accurate (MP2 and CCSD(T) complete basis set limit) interaction energies of small model complexes, DNA base pairs, and amino acid pairs. *Phys. Chem. Chem. Phys.* **8,** 1985–1993 (2006).

[73] Takatani, T., Hohenstein, E. G., Malagoli, M., Marshall, M. S. & Sherrill, C. D. Basis set consistent revision of the S22 test set of noncovalent interaction energies. *J. Chem. Phys.* **132,** 144104 (2010).

[74] Lebeda, T. & Kümmel, S. Meta-GGA that describes weak interactions in addition to bond energies and band gaps. *Phys. Rev. B* **111,** 155133 (2025).

[75] He, J., Hummer, K. & Franchini, C. Stacking effects on the electronic and optical properties of bilayer transition metal dichalcogenides MoS_2, $MoSe_2$, WS_2, and WSe_2. *Phys. Rev. B* **89,** 075409 (2014).

[76] Ferri, N., DiStasio, R. A., Ambrosetti, A., Car, R. & Tkatchenko, A. Electronic Properties of Molecules and Surfaces with a Self-Consistent Interatomic van der Waals Density Functional. *Phys. Rev. Lett.* **114,** 176802 (2015).

[77] Bronsema, K. D., De Boer, J. L. & Jellinek, F. On the structure of molybdenum diselenide and disulfide. *Z. Anorg. Allg. Chem.* **540,** 15–17 (1986).

[78] Borlido, P. et al. Large-Scale Benchmark of Exchange–Correlation Functionals for the Determination of Electronic Band Gaps of Solids. *J. Chem. Theory Comput.* **15,** 5069–5079 (2019).

[79] Mak, K. F., Lee, C., Hone, J., Shan, J. & Heinz, T. F. Atomically Thin MoS_2: A New Direct-Gap Semiconductor. *Phys. Rev. Lett.* **105,** 136805 (2010).

[80] Wallauer, R., Reimann, J., Armbrust, N., Güdde, J. & Höfer, U. Intervalley scattering in MoS2 imaged by two-photon photoemission with a high-harmonic probe. *Appl. Phys. Lett.* **109,** 162102 (2016).

[81] Zhu, Z. Y., Cheng, Y. C. & Schwingenschlögl, U. Giant spin-orbit-induced spin splitting in two-dimensional transition-metal dichalcogenide semiconductors. *Phys. Rev. B* **84,** 153402 (2011).

[82] Xiao, D., Liu, G.-B., Feng, W., Xu, X. & Yao, W. Coupled Spin and Valley Physics in Monolayers of MoS_2 and Other Group-VI Dichalcogenides. *Phys. Rev. Lett.* **108,** 196802 (2012).

[83] Jin, W. et al. Direct Measurement of the Thickness-Dependent Electronic Band Structure of MoS_2 Using Angle-Resolved Photoemission Spectroscopy. *Phys. Rev. Lett.* **111,** 106801 (2013).

[84] Yeh, P.-C. et al. Direct Measurement of the Tunable Electronic Structure of Bilayer MoS2 by Interlayer Twist. *Nano Lett.* **16,** 953–959 (2016).

[85] Behler, J. & Parrinello, M. Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces. *Phys. Rev. Lett.* **98,** 146401 (2007).

[86] Bartók, A. P., Payne, M. C., Kondor, R. & Csányi, G. Gaussian Approximation Potentials: The Accuracy of Quantum Mechanics, without the Electrons. *Phys. Rev. Lett.* **104,** 136403 (2010).

[87] Deng, B. et al. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nat. Mach. Intell.* **5,** 1031–1041 (2023).

[88] Barros-Luque, L. et al. The Open Materials 2024 (OMat24) inorganic materials dataset and models. *Nature Computational Science* **6,** 642–652 (2026).

[89] Kaplan, A. D. et al. A Foundational Potential Energy Surface Dataset for Materials. Preprint at https://arxiv.org/abs/2503.04070 (2025).

[90] Kuner, M. C., Kaplan, A. D., Persson, K. A., Asta, M. & Chrzan, D. C. MP-ALOE: an r2SCAN dataset for universal machine learning interatomic potentials. *npj Comput. Mater.* **11,** 352 (2025).

[91] Schlipf, M. & Gygi, F. Optimization algorithm for the generation of ONCV pseudopotentials. *Comput. Phys. Commun.* **196,** 36–44 (2015).